

Clitic placement and ellipsis

Andrés Saab

Consejo Nacional de Investigaciones Científicas y Técnicas (Conicet)

University of Buenos Aires

andres.saab@uba.ar

Abstract: The theme of this chapter is the interaction between clitic placement and ellipsis. Many case studies are reviewed with the aim of adjudicating among the two main theories of clitics, namely, the Syntax Only and the PF movement approaches. According to the first view, syntax is enough to account for the distributional peculiarities of special clitics. In contrast, according to the PF movement view (and other PF alternatives), syntax alone cannot explain those peculiarities and further morphological or phonological manipulations are also needed. Four case studies are discussed: (i) the distribution of verbal ellipsis and clitic placement in Romance, (ii) the correlation between clitic climbing and the ban of modal ellipsis in Spanish, (iii) the ban of particle reduplication under verbal ellipsis in Hungarian, and (iv) the behavior of clitic clusters in a variety of ellipsis environments in Serbo-Croatian. The entire set of facts requires a theory of clitic placement and clitic clustering in which both syntax and morphology play a crucial role in the derivational histories of clitics.

Keywords: ellipsis, clitics, restructuring, second position, morphology, syntax

1. Introduction

The theme of this chapter is the interaction between clitic placement and ellipsis. There are several reasons why such interaction is scientifically substantive. As is well known, in the (even typologically distant) languages that have them, clitics have many perplexing properties. For instance, in Romance languages, pronominal clitics are syntactically special in two ways. On the one hand, at least superficially, they have a positional behavior that sets them apart both from phrases and affixes. Thus, in Spanish, and in related Romance languages, the accusative clitic occurs as a proclitic, adjoined to the main verb or the auxiliary, when there is one.

- (1) a. Ana leyó el libro.
Ana read the book
'Ana read the book.'
- b. Ana lo leyó.
Ana CL.ACC.M.SG read
'Ana read it.' (lit. 'Ana it read.')
- c. Ana había leído el libro.
Ana had read the book
'Ana had read the book.'
- d. Ana lo había leído.
Ana CL.ACC.M.SG had read
'Ana had read it.' (lit. 'Ana had read it.')

On the other hand, unlike affixes, clitics have some mobility in the clause, sometimes occurring as a proclitic and some other times as an enclitic.

- (2) a. Ana puede leerlo.
Ana can read.INF- CL.ACC.M.SG
'Ana can read it.'

- b. Ana lo puede leer.
 Ana CL.ACC.M.SG can read.INF
 ‘Ana can read it.’ (lit. ‘Ana it can read.’)

A very prominent analysis says that clitics move as phrases in such a way that the proclitic positions in (1b), (1d) and (2b) are the byproduct of a movement operation to some position in the middle field of the clause (Kayne 1975). Yet, the movement analysis is challenged by the fact that many languages have clitic doubling. In Rioplatense Spanish, even lexical direct objects can be doubled by a clitic:

- (3) Ana la vio a Paula.
 Ana CL.ACC.F.SG saw DOM Paula
 ‘Ana saw Paula.’ (lit. ‘Ana her saw to Paula.’)

Clitic doubling not only challenges a simplistic version of the movement theory of clitics, but it also opens new questions regarding the syntactic-semantic nature of clitics and their morphological profile (see Anagnostopoulou 2014 for a detailed overview on clitic doubling).

And there are still more perplexing properties. For instance, regardless of the nature of clitic movement, clitics use to move as clusters. Concretely, whenever there is more than one clitic in the same clausal domain, they climb together:

- (4) a. Ana puede dármelo.
 Ana can can.INF.CL.DAT.1SG-CL.ACC.M.3SG
 ‘Ana can give it to me.’
 b. Ana me lo puede dar.
 Ana CL.DAT.1SG CL.ACC.M.3SG can can.INF
 ‘Ana can give it to me.’ (lit. ‘Ana me it can give.’)
 c. * Ana me puede darlo.
 Ana CL.DAT.1SG can can.INF-CL.ACC.M.3SG
 d. * Ana lo puede darme.
 Ana CL.ACC.M.3SG can can.INF-CL.DAT.1SG

Similar considerations regarding the peculiar behavior of pronominal clitics in Romance also arise around clitic second-position, in languages like, for instance, Serbo-Croatian and Slavic languages in general. In effect, although this language is extremely flexible with respect to word ordering, compared to other types of constituents, clitics are much more rigid, in the sense that they are found in the second position of their matrix sentence, i.e., either after the first phrase (5a) or after the first word (5b):

- (5) a. [Tu knjigu] su mi dali.
 that book.ACC are me-DAT given
 ‘They gave that book to me.’
 b. [Tu] su mi knjigu dali.
 that are me-DAT book.ACC given
 ‘They gave that book to me.’

[Stjepanović 1999: 14]

And like what I have just observed with respect to pronominal clitics in Romance, second-position clitics in Serbo-Croatian also form clusters. Thus, in part because of all these distributional peculiarities, pronominal and second-position clitics are called “special clitics” in the terminology of Zwicky (1977). Mainstream theories of special clitics in the generative framework adopted in this chapter are divided between purely syntactic analyses and hybrid approaches that conceive of clitics as having a more complex derivational history between proper syntax and morphology. Embick and Noyer (2001) state the debate in the following terms:

“*Syntax only*: Syntax performs operations that are explicitly executed so as to resolve a morphophonological problem. Patterns of apparent postsyntactic movement are reducible to the effects of these “special” syntactic processes.

PF movement: Syntax generates and moves terminals according to its own principles and is oblivious to morphophonological concerns. PF takes the output of syntax and resolves morphophonological dependencies according to its own principles.”

[Embick and Noyer 2001: 556-557]

Part of the controversy is, then, whether syntax is the sole responsible for clitic placement and the perplexing behavior of clitics (*Syntax only*) or whether further morphological manipulations also play a role in that behavior (*PF movement*).

Stjepanović (1999) made the same point regarding second-position clitics in Serbo-Croatian and related languages:

“The major issues debated with respect to the second position cliticization puzzle focus on whether the placement of clitics in the second position is achieved exclusively by exploiting movement in syntax, or whether some postsyntactic word reordering is necessary as well.”

[Stjepanović 1999: 13]

In addition, she also made the important claim that ellipsis is a powerful grammatical tool to adjudicate between the *Syntax Only* view and alternative post-syntactic movement approaches. In fact, she offered one of the first in-depth studies on the interaction between clitic placement and ellipsis (see section 5). The idea is based on the very well-known fact that ellipsis can bleed morphological operations. A famous case is affix hopping in English, which is blocked under VP-ellipsis. As discussed originally by Chomsky (1957), when this happens, the morphophonology of English inserts the pleonastic element *do* to support the stranded inflectional affix.

(6) Blocking affix hopping by ellipsis:

a. Anne worked hard and I *-ed* [_{VP} work hard] too.

Do-support:

b. ...and I *did* too.

In the same vein, Lasnik (1995) used this type of facts in English to assess different theories of T⁰ to V⁰ affixation in English which were in competition in the first minimalist works by Chomsky and others. As discussed at length in Saab (2008) and Saab and Lipták (2016), this kind of ellipsis/morphology interaction is not a fortuitous fact of the English grammar but generalizes cross-linguistically. Language after language, ellipsis blocks certain types of post-syntactic operations, feeding in some cases the application of support

strategies (although see Georgevia *et al* 2020 for a putative counterargument). As is clear, well controlled ellipsis environments should inform the two theories of clitics under dispute. The case studies that follow show that syntax cannot be the sole responsible for clitic placement. As argued under the PF Movement approach, further morphological manipulations play a crucial role in the derivation of clitics cross-linguistically. But before discussing those case studies, in section 2, I first introduce a set of important assumptions about clitics and ellipsis, which will be relevant for framing the subsequent discussion. Then, in section 3, I offer the first case study on ellipsis and clitic placement in Romance. Concretely, I critically review Martins’ (1994) generalization according to which enclisis in finite clauses correlates with the licensing of verbal ellipsis in Romance. Based on a previous study in Saab (2022a), I argue that, in reality, the correlation is not exactly about clitic placement and ellipsis, but between ellipsis and PF restructuring, an operation that combines T^0 and V^0 at PF. Languages with PF restructuring have two crucial properties: (i) lack of VP-ellipsis (and VP-movement), and (ii) have uniform proclisis in finite clauses. In section 4, I present independent evidence from canonical restructuring predicates in Spanish and from particle reduplication in Hungarian. In section 5, I review Stjepanović’s (1999) study on clitic second position and ellipsis in Serbo-Croatian. This language constitutes an ideal scenario to test different approaches to clitic placement and clitic clustering, since the language has a variety of verbal ellipses with stranded clitics. In the concluding section, I summarize the main results and briefly comment on other types of relevant interactions between clitics and ellipsis.

2. Assumptions

The discussion that follows shares the research agenda and the main assumptions of Distributed Morphology (DM; Halle and Marantz 1993, 1994, Embick and Noyer 2001, Saab 2008, Arregi and Nevins 2012, and Harley 2014, among many others). Different versions of DM adopt the following architecture of grammar:

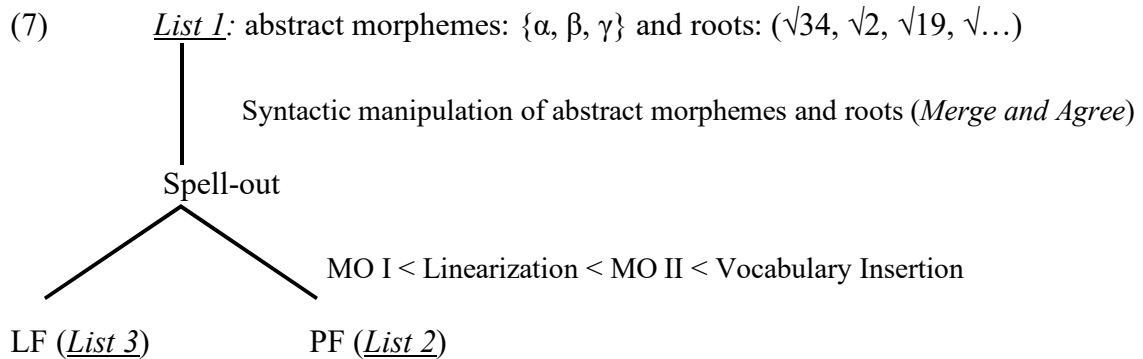


Figure 1. *Architecture of grammar (MO = Morphological Operation)*

A core DM idea is that there is no single lexical module in which word formation takes place. Indeed, “word” is just a descriptive label for different formal objects built out of an initial array of roots and abstracts morphemes (List 1 in Figure 1) and a set of grammatical principles of morphosyntactic combination taking place in the path from syntax to PF. In the version of the model adopted here, both roots and abstracts morphemes have no phonological information in List 1. At PF, a set of Vocabulary Insertion rules provides the vocabulary items with phonological information on the abstract terminal nodes. Many morphological manipulations at PF can alter the output of syntax before Vocabulary Insertion. These manipulations mainly consist of a variety of morphological mergers (varieties of Lowering and Local Dislocation in Embick and Noyer’s (2001) model), feature copying, dissociated node insertion (case and agreement

nodes, among other notions) and morphological “deletions” subject to the same locality principles as other morphological operations (Saab 2008). In this section, I introduce my main assumptions regarding clitics and ellipsis in order to explicitly frame the subsequent discussion.

2.1. Clitics

Broadly speaking, most theories of clitics divide into two families. On the one hand, there are those who think that clitics are a special type of morphosyntactic entity, different from phrases and affixes (see, in particular, Zwicky 1977, 1985 and Zwicky and Pullum 1983). On the other hand, clitics can be conceived not as some sort of primitive, but as the result of the way in which grammar builds phrases and heads out from an initial array of lexical primitives. What is special about clitics on this view is their hybrid phrase structure behavior. The crucial observation is that they can behave simultaneously as phrases or heads. In the bare phrase structure framework, Chomsky captures such hybrid behavior by relativizing levels of projections, in such a way that, by definition, there is no inconsistency for a category to count as minimal and maximal under some conditions (Chomsky 1995a,b). Yet, a more explicit criticism of the primitive distinction between, say, affixes and clitics is found in Embick and Noyer (2001) in the Distributed Morphology framework (Halle and Marantz 1993, 1994 and much subsequent work). In their own words:

“In the context of the present theory, the modular distinction between clitics and affixes cannot be maintained. That is, there is no Lexical mode of derivation in the present theory, so such a distinction loses much of its importance. Instead, elements that appear bound to others on the surface have heterogeneous derivational histories, with *clitic* and *affix* being descriptive terms for some of these. The important factor in any particular analysis is therefore not whether an element is a clitic or an affix. Rather, one must identify (a) the provenance of the morpheme (syntactic or dissociated) and its distribution, and, correspondingly, (b) the means by which it comes to be attached to its host.”

[Embick and Noyer 2001: 560, my underling]

In this chapter, I will be aligned with Embick and Noyer’s idea that clitics and affixes are not primitive ingredients of morphosyntactic theory. Any syntactic derivation starts with an initial array of items taken from a list of abstract morphemes and roots (List 1 in Figure 1). Syntactic manipulations produce two types of structural objects: phrases and heads. Yet, under the bare phrase structure approach, such a distinction is not perfectly transparent and requires additional comments. As said above, Chomsky (1995a,b) introduced the labels *min* and *max* to distinguish the basic levels of projections but adds that more distinctions are needed at the head level. Complex heads, for instance, are of some special sort and he proposed the label H^{0max} to distinguish them from pure terminal nodes and maximal phrases. Embick and Noyer (2001) make a similar distinction through the notions of *morphosyntactic word* and *subword*, defined as follows:¹

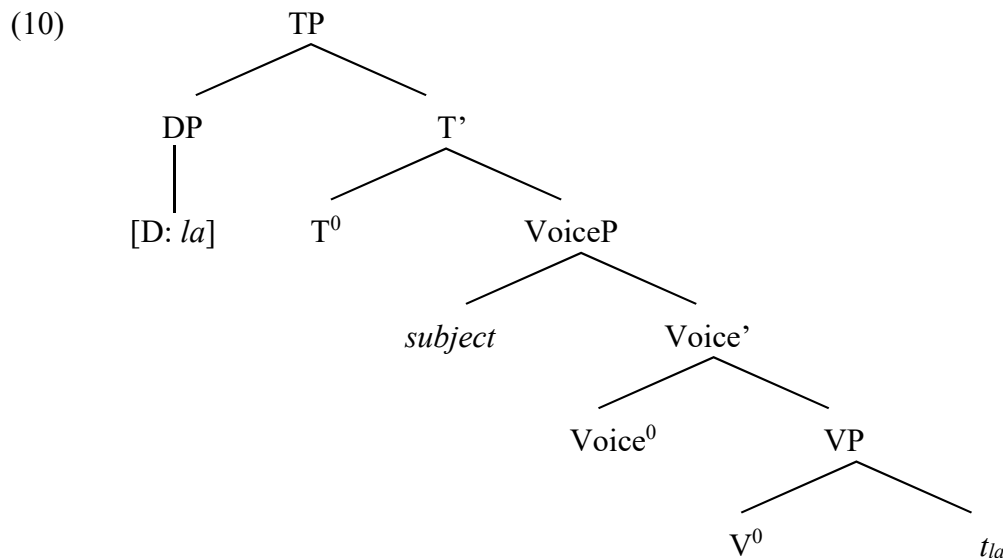
- (8) a. *Morphosyntactic word*: At the input to Morphology, a node X^0 is (by definition) a *morphosyntactic word* (MW) iff X^0 is the highest segment of an X^0 not contained in another X^0 .
- b. *Subword*: A node X^0 is a *subword* (SW) if X^0 is a terminal node and not an MW.

¹ Through the chapter, I use the superscript ⁰ for MWs, and no superscript for SWs,

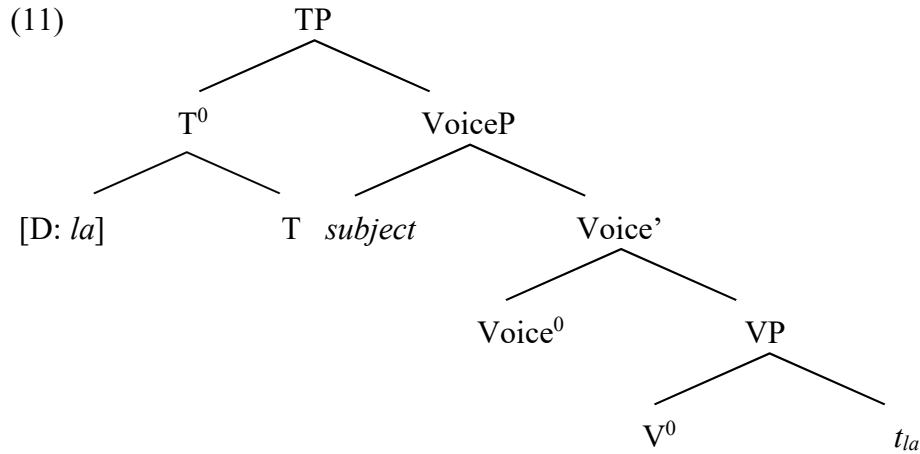
Now, the phrasal status of any given syntactic object can be altered in the way from syntax to PF. An important assumption is that the hybrid phrasal nature of clitics is due to the way in which PF manipulates the output syntax produces. In DM, these manipulations are, among other formal operations, PF movements subject to morphological conditions. Lowering and Local Dislocation are the two movement operations discussed in Embick and Noyer (2001). On this view, (at least some type of) clitics are just syntactic phrases that become heads at PF as the byproduct of some of these movement operations. Take, for instance, pronominal clitics of the Romance type.

- (9) Ana la vio.
 Ana CL.ACC.F.3SG saw
 ‘Ana saw her.’

Derivationally, one can assume that the accusative clitic is generated as a maximal phrase in the complement position of the verb. Then, it moves in the syntax to some position in the middle field of the clause. This movement is just movement from one phrase position to another phrase position. At PF, an additional operation adjoins the clitic to T^0 or to some related functional category. This is essentially the analysis in Matushansky (2006: 85). The syntactic step is given in (10), in which the clitic moves from its complement position to Spec,TP (although this is not crucial, it could be another lower position, as well):



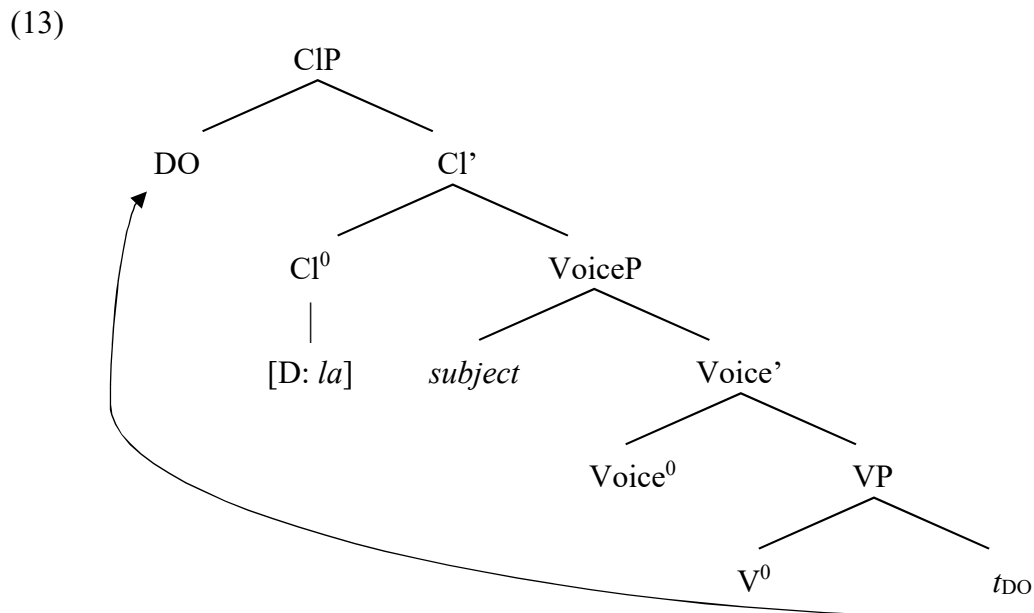
At PF, Matushansky proposes that the clitic adjoins by M-Merger (Local Dislocation in Embick and Noyer's model) giving rise to something closer to the final string:



Now, clitic doubling of the type illustrated in (3) must have a different structural source:

- (12) Ana la vio a Paula.
 Ana CL.ACC.F.3SG saw DOM Paula
 ‘Ana saw Paula.’

There are many different analyses of clitic doubling for Spanish and across languages. With Di Tullio *et al* (2019) and Saab (2022b, 2024b), I assume that doubling clitics are probes for A-movement. Therefore, unlike argument clitics, doubling ones are base-generated as probes in the middle field of the clause, position from which they trigger A-movement of the doubled object. Schematically, this is illustrated in (13):



So far, the hybrid nature of clitics is explained as the byproduct of their complex derivational history from syntax to morphology. Different clitic configurations (e.g., clitic doubling vs. simple clitic configurations) have a different structural source with clitics sometimes merged as arguments and some other times as functional material above the clausal lexical domain (although see the concluding section for a more uniform

alternative). From a descriptive point of view, we have as a minimum a bipartite division between clitics as pronouns or agreement probes. Against this pure bipartite division or the one proposed in Baker and Kramer (2018), there is still room for another derivational object, namely, clitics as agreement markers, i.e., pure set of ϕ -features (see Saab 2022b, 2024b). I will not be concerned with the proper analysis of each type of clitic. Again, the core idea is that there are not primitive types, but a set of different derivations for abstract morphemes and the combinations that result from them.

2.2. Ellipsis

Ellipsis is relevant to the theory of clitics because of the type of interaction observed between different kinds of ellipsis and the derivation of clitics. In effect, clitic placement is affected by ellipsis in ways that shed light on the theory of clitics sketched in the previous section. To start with, I will assume that ellipsis is an operation of core grammar that applies at different derivational steps from narrow syntax to PF (Saab 2008, 2022c). When it applies in the syntax, ellipsis affects phrases under well-known syntactic conditions (c-command or selection), whereas when it applies at PF, it targets morphosyntactic words, as defined in (8a). Head ellipsis is constrained by morphological locality conditions, in principle, the same relevant for different types of morphological movements namely, immediate locality and adjacency. Which locality condition is relevant depends on linearization. In this sense, like Lowering, when head ellipsis takes place before linearization, it obeys immediate locality, which is the relation between a head and the head of its complement. In turn, when the elision applies after linearization, it obeys the same locality condition as Local Dislocation, i.e., adjacency. Having said that, in this chapter, I will focus mainly on syntactic ellipsis, although I will also offer some brief comments on head ellipsis in the concluding section. The architecture for this model of derivational ellipsis is illustrated with the scheme in (14):

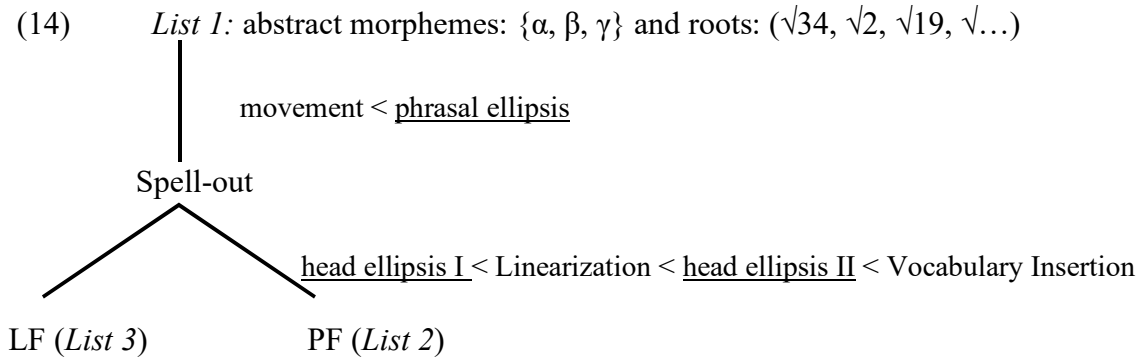
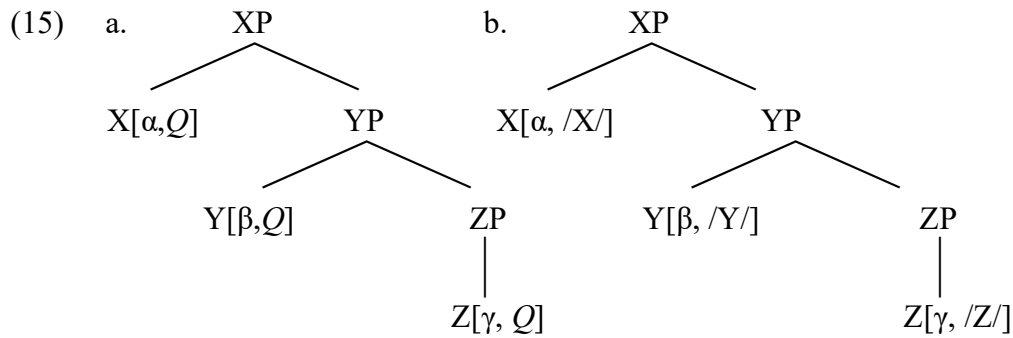


Figure 2. *Derivational ellipsis*

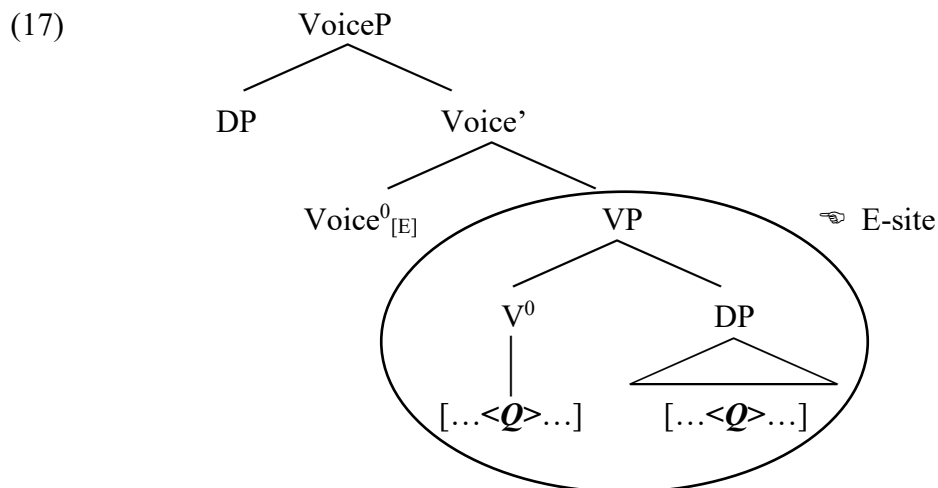
Another important assumption is that ellipsis is not a PF-deletion operation, but a mechanism of vocabulary insertion blocking. This is implemented as deletion of the syntactic features that trigger vocabulary insertion, the [Q]-features in Halle (1991) and Embick (2015). On this replacive view of Vocabulary Insertion, syntax provides each terminal node with a [Q]-feature, which is essentially read as a free variable replaced by a vocabulary item at PF. Roughly, the process of Vocabulary Insertion consists of *Q*-replacement, as illustrated below:



In Saab (2022c), ellipsis is seen as Q -deletion. As illustration, consider as case of verbal ellipsis in English ($\langle \dots \rangle = \text{E-site}$), which I assume is ellipsis of the complement of Voice (Merchant 2013):

(16) Anne likes movies and Pauline does $\langle [\text{VP like movies}] \rangle$ too.

The Voice head licenses ellipsis of its complement whenever it encodes a $[E]$ -feature in Merchant's (2001, 2004) sense. If the VP selected by $\text{Voice}_{[E]}$ has a lexical and syntactically identical linguistic antecedent, then ellipsis deletes the $[Q]$ -features contained in each terminal node of the VP projection. This is illustrated as follows ($\langle Q \rangle = Q$ -deletion):



In summary, both phrases and morphosyntactic words can be subject to Q -deletion under different formal conditions. An important property of this model is that parts of words, i.e., subwords as defined in (8b), cannot be elided with independence of the morphosyntactic word dominating them. This is called the *Subword Deletion Corollary* in Saab (2008) and subsequent work, and I will assume it in what follows.

With these assumptions in mind, we can now see why ellipsis is particularly relevant to the morphosyntactic derivation of clitic placement. Recall that I have assumed that clitics do not exist as primitives of grammatical theory. They are complex hybrid objects, whose particular status and behavior are explained by the application of different syntactic and morphological principles of phrase and word formation. For instance, I assumed that pronominal clitics are merged as phrases of their selector predicate and moves to some phrasal position in the middle field of the clause. A PF rule, then, converts the clitic into a subword of its host (Matushansky 2006). Ellipsis intervenes in this type of derivational history by, for instance, eliminating a potential clitic host or by destroying the

environment for clitic adjunction. This is a consequence of the following generalization (Saab 2008 and Saab and Lipták 2016, among others):

- (18) *Ellipsis-Morphology Generalization (Elmo)*: For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

In the following sections, we will see several case studies where the Elmo generalization is operative, explaining part of the behavior of clitic placement under ellipsis and illuminating in particular ways the general theory of clitic placement and clitic clustering. In principle, the evidence points out in favor of the derivational theory of special clitics roughly sketched here.

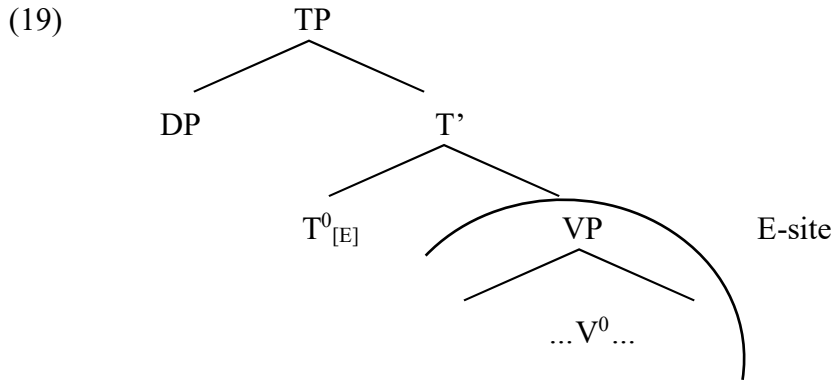
3. Clitic placement and verbal ellipsis in Romance

In this section, I address the interaction between clitic placement and verbal ellipsis in Romance. Concretely, I review Martins' (1994) generalization, according to which verbal ellipsis correlates with certain type of enclisis in finite clauses. This predicts verbal ellipsis in enclitic languages like Galician and European Portuguese but not in uniform proclitic languages like Spanish, French, Catalan or Italian.² Martins accounts for this observation by the scope of head movement in each Romance language and the licensing of ellipsis, which, according to her, is due to the activity of the polarity head, Σ^0 (Laka 1990). In Galician or European Portuguese, the verb moves to Σ^0 , giving rise to enclisis and the licensing of ellipsis. Here, I offer both a criticism of Martins's proposal and an analytical alternative, which reduces uniform proclisis in finite tenses to clitic climbing. Clitic climbing is prevented under ellipsis because it bleeds PF restructuring, a precondition for clitic climbing (Rizzi 1982). Thus, this section offers the first case study in favor of the theory of clitics and ellipsis introduced in the previous section. But before advancing in the relevant empirical domain, I first discuss the type of verbal ellipses I will be concerned with, namely, *auxiliary and verb-stranding VP-ellipsis*.

3.1. Two varieties of verbal ellipsis across languages

VP-ellipsis is defined as a type of surface anaphora (Hankamer and Sag 1976) in which the E-site corresponds to some projection of the lexical verbal domain (e.g., VoiceP, VP, etc.). There is certain consensus that VP-ellipsis requires licensing by inflection. In terms of Merchant's theory, licensing is implemented through the presence of a formal feature, [E](llipsis), which essentially instructs deletion/non-pronunciation of the complement of whatever the head that encodes the relevant [E]-feature. On this view, if T^0 encodes [E], then VP (VP = some projection of the verbal domain) is elided provided that some identity conditions between the E-site and a suitable linguistic antecedent is met. Roughly, we can illustrate the basic ingredients of VP-ellipsis as follows:

² By *uniform proclisis*, I refer to the fact that these languages, unlike Galician, Portuguese and other Romance varieties, do not make any difference between, say, negative or affirmative sentences: proclisis is attested all over across indicative or subjunctive mood. This property is not trivial for the argument to be developed, since it is an indication of the more rigid behavior of clitics in the relevant languages.



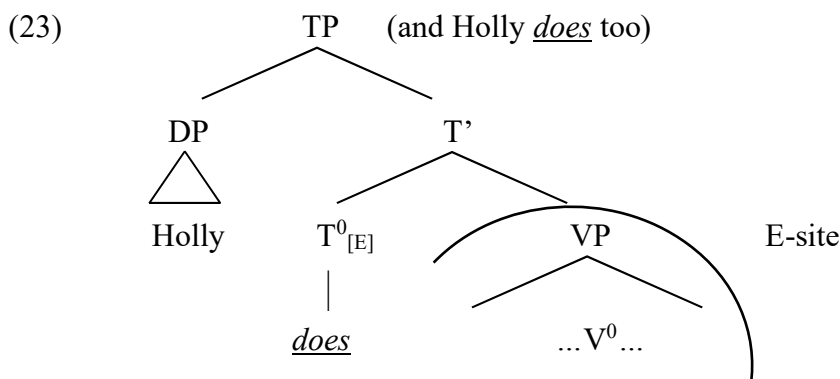
English is the most studied VP-ellipsis language. Specifically, the language shows a variety of ellipsis known as *auxiliary*-stranding VP-ellipsis, a type of verbal ellipsis in which the E-site is preceded by a stranded auxiliary (20) or negation (21):

- (20) a. José Ybarra-Jaegger likes rutabagas, and Holly does too.
 b. José Ybarra-Jaegger ate rutabagas, and Holly has too.
 c. José Ybarra-Jaegger should have eaten rutabagas, and Holly should have too.
 d. José Ybarra-Jaegger is eating rutabagas, and Holly is too.
 e. José Ybarra-Jaegger has been eating rutabagas, and Holly has been too.
 d. Mag Wildwood wants to read Fred's story, and I also want to.
 [Johnson 2001: 440]
- (21) a. John is leaving but Mary's not.
 b. I consider Bill intelligent and I consider Sally not.
 [Johnson 2001: 440]

As is well-known, English is a language in which T^0 lowers to V^0 at PF under a strict locality condition (adjacency or immediate locality, Chomsky 1957, Lasnik 1995, Bobaljik 1995, and Embick and Noyer 2001, among many others). Now, verbal ellipsis bleeds lowering, consequently leaving the T^0 node stranded instantiating a concrete case of the *Ellipsis-Morphology Generalization*:

- (22) *Ellipsis-Morphology Generalization (Elmo)*: For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

In those cases in which T^0 is morphologically realized as an affix, like in (20a), a violation of the Stranded Affix Filter obtains (Lasnik 1981). The language repairs the stranded affix scenario by providing a verbal host to the T^0 node. For a sentence like (20a), the relevant analysis is provided in the tree below, which illustrates VP-ellipsis and *do*-insertion:

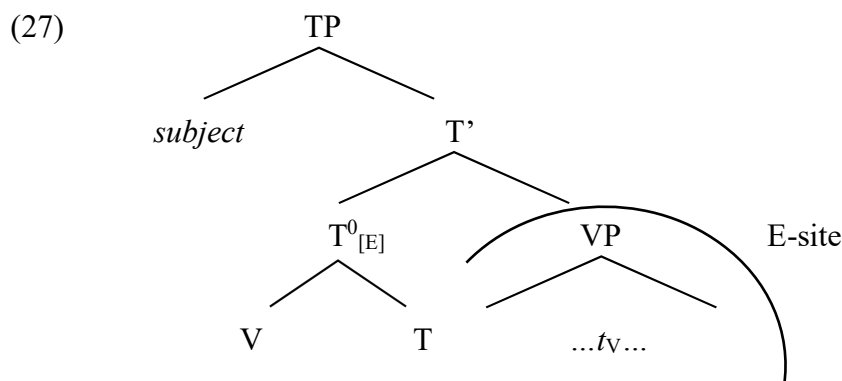


(24) Portuguese:
a. Quando a Ana pôs os óculos na mesa,
when the A. put the glasses on-the table,
a Maria também pôs < os óculos na mesa>.
the Maria too put the glasses on-the table.
‘When Ana put the glasses on the table, Maria did too.’
[Cyrino and Matos 2002: 6]
b. O Luís foi à biblioteca às nove horas
the Luís went to.the library at nine o’clock
e o Pedro também foi < ____ >.
and the P. too went
‘Luís went to the library at nine o’clock and Peter did, too’.
[Cyrino and Matos 2005: 9]

(25) Irish:
A: A-r bhain tú sult as?
INTERR[PAST] take you fun from-it
A: ‘Did you enjoy it?’
B: Bhain.
took
‘I did.’ (Lit. ‘I took.’)
[McCloskey 2004]

(26) Hebrew:
[Context: Dvora is pregnant and has many errands to do; Miryam, who has a car but is sometimes inconsiderate, is supposed to be helping her.]
Q: (Ha'im) Miryam hevi'a et Dvora la-xanut?
Q Miryam brought.F.3SG ACC Dvora to.the-store
‘(Did) Miryam bring Dvora to the store?’
A: Ken, hi hevi'a.
yes she brought.F.3SG
‘Yes, she brought <Dvora to the store>.’
[Goldberg 2005: 160]

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Yet not all languages for which V-to-T movement has been proposed license verbal ellipsis, raising the question of why this is the case. Consider Spanish, which has neither Aux- nor V-stranding VP-ellipsis, but which, according to most views, does have V⁰-to-T⁰ movement:

- (28)
- Aux-stranding VP-ellipsis: no
- a. A: Habías comprado el libro?
 had bought.2SG the book
 ‘Had you bought the book?’
- B: * Sí, HABÍA.
 yes had.1SG
 Intended: ‘Yes, I had.’
- V-stranding VP-ellipsis: no
- b. A: Le diste el libro?
 CL.DAT.3SG gave.2SG the book
 ‘Did you give her/him the book?’
- B: * Sí, DI.
 yes gave.1SG
 Intended: ‘Yes, I did.’ (lit. ‘Yes, I gave.’)

The few attempts at offering a minimally reasonable answer to the problem of the VP-ellipsis parameter have been unsuccessful. One of the most articulated theories of ellipsis licensing is that of Lobeck (1995), who proposes to correlate the presence or absence of verbal ellipsis to the timing of verbal head movement. On her account, Spanish, which has overt head movement, does not admit verbal ellipsis, but English, which does not have verbal movement, does license verbal ellipsis. The problem with Lobeck’s account is that there are languages with V-stranding ellipsis, which obviously falsify her expectations. Zagana’s (1982, 1988) attempts have not been more successful, although many of the thoughts I will offer here follow many of her original insights. As already commented, in current minimalism, the best-known implementation of ellipsis licensing is Merchant (2001, 2004), according to whom ellipsis is licensed by the presence of a formal feature, his [E(ellipsis)]-feature. More concretely, Merchant, elaborating on Lobeck (1995) and, more generally, on the Borer/Chomsky’s conjecture, suggests that we should encode the parameter in lexical properties of (mainly) functional heads. Yet, under this approach there is no expectation for this state of affairs: since the placement of the [E]-feature depends on lexical options of particular languages, we should expect a random distribution of ellipsis sizes across languages (see Thoms 2012 for a very similar criticism). But this is not the general case. For instance, whereas fragments or sluicing are well-attested in a great number of languages, VP-ellipsis is only attested in a reduced set of languages. But there is more. One of the main problems for the [E]-feature approach

is that VP-ellipsis is not entirely casual. In this respect, the interaction between clitic placement and ellipsis in Romance constitutes an ideal scenario for parametric investigation.

3.2. VP-ellipsis, head movement and clitic placement in Romance

In the Romance realm, Martins (1994) made the important observation that languages with productive enclisis in finite sentences concomitantly license VP-ellipsis both of the Aux- and of the V-stranding type.³ I state this in the following way:

- (29) Martins' Generalization: If a Romance language has enclisis as the byproduct of syntactic head movement, then the language has also V-stranding VP-ellipsis.

The following patterns show that this seems to be correct for both varieties of verbal ellipses introduced in the previous section, namely Aux- and V-stranding VP-ellipsis (small capitals indicate polarity focus):

(30) *European Portuguese*

Aux-stranding VP-ellipsis: yes

- a. A. Você tinha dado o livro pra Sonia?
 you had.1SG given the book to.the Sonia
 'Did you give the book to Sonia?'
 B. Sim, TINHA.
 yes had.1SG
 'Yes, I had.'

V-stranding VP-ellipsis: yes

- b. A. Deste-lhe o livro?
 gave.2SG-CL.DAT.3SG the book
 'Did you give him/her the book?'
 B: Sim, DEI.
 yes, gave.1SG
 'Yes, I did.'

Enclisis: yes

- c. A: Deste-lhe o livro?
 gave.2SG-CL.DAT.3SG the book
 'Did you give him/her the book?'
 B: Sim, DEI-LHO.
 yes, gave.1SG-CL.DAT.3SG.CL.ACC.3SG
 'Yes, I gave it to him/her.'

(31) *Galician*

Aux-stranding VP-ellipsis: yes

- a. A: Vostede tínhalle dado o libro a Ana?

³ I am assuming with Martins that these languages have true verbal ellipsis of some projection of the VoiceP domain, but it could be also the case that the relevant languages license a higher type of V-stranding ellipsis, e.g., V-stranding TP-ellipsis. Nothing in Martins's and the argumentation in this section change if this is the case and the relevant languages do not have V-stranding VP-ellipsis but a higher type of verb stranding ellipsis. See Landau (2020a,b) for arguments against V-stranding VP-ellipsis but in favor of higher XP-ellipses, and Saab (2022a) for some counterarguments.

you had.2SG-CL.DAT.3SG given the book to Ana
 ‘Did you give the book to Ana?’

B: Si, TIÑA.
 yes had.1SG
 ‘Yes, I had.’

V-stranding VP-ellipsis: yes

b. A: Décheslles o libro?
 gave.2SG- CL.DAT.3SGthe book
 ‘Did you give him/her the book?’

B: Si, DIN.
 yes, gave.1SG
 ‘Yes, I did.’

Enclisis: yes

c. A: Décheslles o libro?
 gave.2SG- CL.DAT.3SGthe book
 ‘Did you give him/her the book?’

B: Si, DINLHO.
 yes, gave.1SG-CL.DAT.3SG.CL.ACC.3SG
 ‘Yes, I gave it to him/her.’

(32) *Spanish*

Aux-stranding VP-ellipsis: no

a. A: Habías comprado el libro?
 had bought.2SG the book
 ‘Had you bought the book?’

B: * Sí, HABÍA.
 yes had.1sg
 Intended: ‘Yes, I had.’

V-stranding VP-ellipsis: no

b. A: Le diste el libro?
 CL.DAT.3SG gave.2SG the book
 ‘Did you give her/him the book?’

B: * Sí, DI.
 yes gave.1SG
 Intended: ‘Yes, I did.’ (lit. ‘Yes, I gave.’)

Proclisis: sí

c. A: Le diste el libro?
 CL.DAT.3SG gave.2SG the book
 ‘Did you give her/him the book?’

B: Sí, se lo di.
 yes CL.DAT.3SG CL.ACC.3SG gave.1SG
 ‘Yes, I gave it to him/her.’

(33) *Catalan*

Aux-stranding VP-ellipsis: no

a. A: Li has donat el llibre?

CL.DAT.3SG have.2SG given the book
 ‘Did you give him/her the book?’

B: * Sí, he donat.
 yes have.1SG given
 Intended: ‘Yes, I have.’

V-stranding VP-ellipsis: no

b. A: Li has donat el llibre?
 CL.DAT.3SG have.2SG given the book
 ‘Did you give him/her the book?’

B: * Si, he donat.
 yes have.1SG given
 Intended: ‘Yes, I did.’ (lit. ‘Yes, I have given.’)

Proclisis: yes

c. A: Li has donat el llibre?
 CL.DAT.3SG have.2SG given the book
 ‘Did you give him/her the book?’
 Si, l’hi he donat.
 yes, CL.ACC.3SG-CL.DAT.3SG have.1SG given
 ‘Yes, I gave it to him/her.’

(34) *French*

Aux-stranding VP-ellipsis:

a. A: Lui as-tu donné le livre?
 CL.DAT.3SG have.2SG-you given the book
 ‘Did you give him/her the book?’

B: * Oui, j’AI.
 yes I-have.1SG
 Intended: ‘Yes, I did.’

V-stranding VP-ellipsis: no

b. A: Lui as-tu donné le livre?
 CL.DAT.3SG have.2SG-you given the book
 ‘Did you give him/her the book?’

B: * Oui, j’ai DONNÉ.
 yes I-have.1SG given
 Intended: ‘Yes, I did.’ (lit. ‘Yes, I have given.’)

Proclisis: yes

c. A: Lui as-tu donné le livre?
 CL.DAT.3SG have.2SG-you given the book
 ‘Did you give him/her the book?’
 B: Oui, je le lui ai donné.
 yes, I CL.ACC.3SG CL.DAT.3SG have.1SG given
 ‘Yes, I gave it to him/her.’

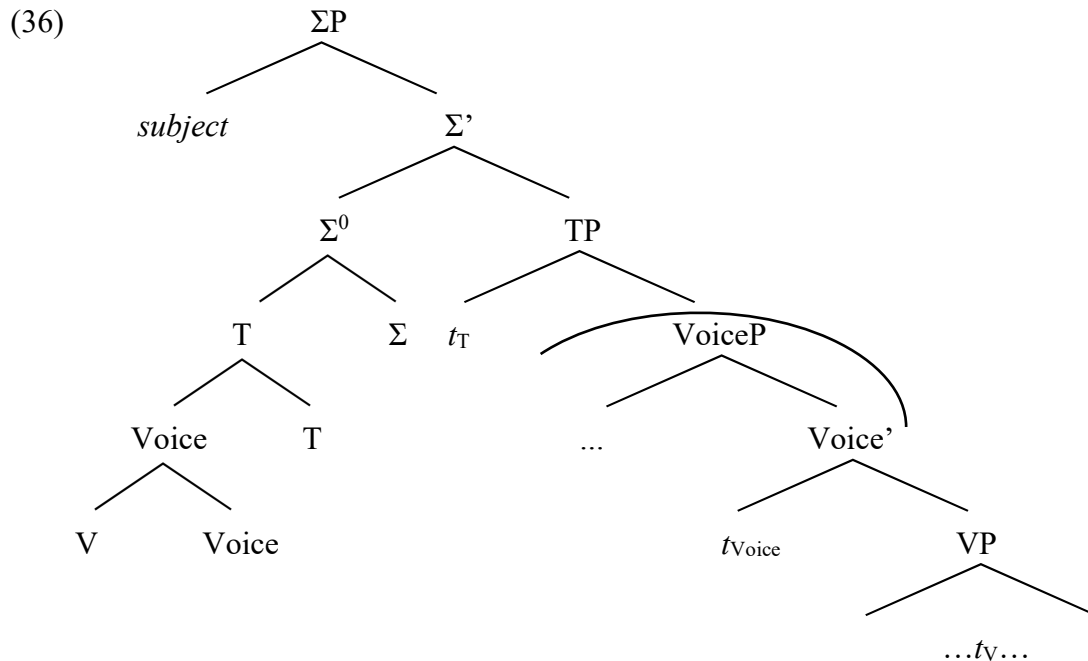
(35) *Italian*

Aux-stranding VP-ellipsis: no

a. A: Gli hai dato il libro?

- CL.DAT.3SG have.2SG given the book
‘Did you give him/her the book?’
- B: * sì, HO.
yes have.1SG
‘Yes, I had.’
- b. A: Gli hai dato il libro?
CL.DAT.3SG have.2SG given the book
‘Did you give him/her the book?’
- B: * sì, ho DATO.
yes have.1SG given
‘Yes, I have.’ (lit. ‘Yes, I have given.’)
- Proclisis: yes
- c. A: Gli hai dato il libro?
CL.DAT.3SG have.2SG given the book
‘Did you give him/her the book?’
- B: Sì, gliel’ ho dato.
yes CL.DAT.3SG.CL.ACC.3SG-have.1SG given
‘Yes, I gave it to him/her.’
- [V-stranding VP-ellipsis from Martins 1994: 174-175]

Martins proposes to explain the correlation in (29) from the length of verbal movement in each language. Adapting Martins’ analysis to more recent conceptions of clause structure and ellipsis, I assume that in those languages in which there is enclisis and ellipsis, the verb moves to Σ^0 (i.e., the functional head that introduces polar values in the sentence) passing cyclically through T^0 . This high verbal movement has two consequences: (i) it gives rise to some instances of enclisis, and (ii) licenses verbal ellipsis. One possible adaptation of Martins’ main insights is shown in the tree below:



Among Romance languages, this type of head movement is particularly productive in Portuguese and Galician (see (30) and (31)), but not in the rest of the Romance languages exemplified in (32)-(35). Yet, Martins (1994) argues that morphological imperatives in

languages like Spanish, of the type exemplified in (37), for which head movement analyses have been proposed (Laka 1990 and Rivero and Terzi 1995), provide independent evidence for her proposal. As is well known, morphological imperatives: (i) require enclisis, and (ii) are incompatible with negation:

- (37) a. Cerrála! vs. *La cerrá!
 close.IMP-CL.ACC.3SG CL.ACC.3SG close.IMP
 ‘Close it!’
 b. No la cierres! vs. *No cerrála!
 not CL.ACC.3SG close.SUBJ not close.SUBJ-CL.ACC.3SG
 ‘Don’t close it!’

For facts like these, several authors have proposed that morphological imperatives move like heads to a position above T^0 . Suppose along the lines of Rivero and Terzi (1995) that the relevant position is, in fact, C^0 itself. According to this analysis, enclisis would be accounted for as an effect of high head movement, which leaves the clitic attached to T^0 behind the verb, while the incompatibility with negation would be explained as an intervention effect, in which the negation blocks verb raising and forces the insertion of a subjunctive verbal form:

- (38) a. [_{CP} cerrá [_{TP} la t]] $\Rightarrow$ *Enclisis*
 b. *[_{CP} cerrá [_{ΣP} no [_{TP} la t]]] $\Rightarrow$ *Neg intervention*

This analysis is in line with the generalization stated in (29). On the basis of examples like (39), Martins (1994) reached the conclusion that imperatives also obey the relevant generalization.

- (39) [Pointing to an open window]
 Cerrá!
 close.IMP
 ‘Close it!’

The problem is that examples like (39) can hardly be conceived of as cases of ellipsis. As Silva (2023) shows, it is simply impossible in Spanish to have V-stranding ellipsis leaving a stranded verb with the morphological imperative form:

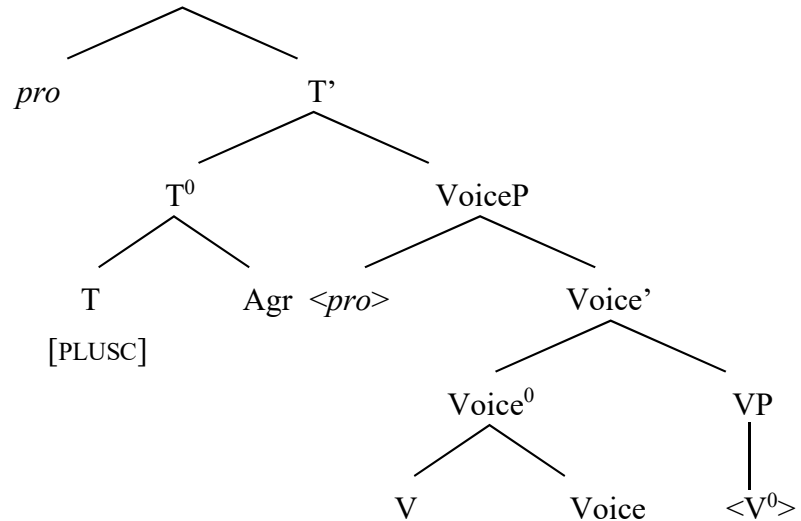
- (40) A: No le compré el regalo a Ana.
 not CL.DAT.3SG bought.1SG the present to Ana
 ‘I didn’t buy the gift to Ana.’
 B: * Por favor, compra < el regalo a Ana >!
 please buy.IMP the present to Ana
 ‘Please, do!’ (Lit. ‘Please, buy (the present to Ana).’)

According to Silva, examples like (39) instantiate cases of exophoric null objects pragmatically controlled. See Silva’s study for more arguments against ellipsis in imperative contexts.

Saab (2022a) offers a different explanation of Martins Generalization, based on the idea that proclisis correlates with T^0+V^0 restructuring. Concretely, Spanish and related languages do not have V^0 to T^0 movement in the syntax but a PF rule that combines T^0

and V^0 in a sense similar to Rizzi's (1982) original restructuring theory. Restructuring applies both in synthetic (e.g., *amaba*) and periphrastic (e.g., *había amado*) tenses, i.e., restructuring is not affixation. As an illustration, consider the basic analysis for the analytic *pluscumperfecto* *habíamos luchado* '(we) had struggled'. In the syntax, there is no V^0 to T^0 movement, and T^0 and the VoiceP stand in the head-complement relation:

(41) Syntax for *habíamos luchado* ➦ TP

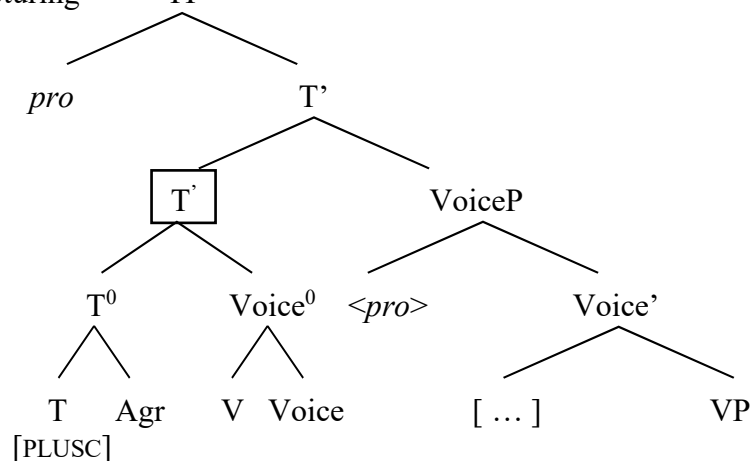


As a PF rule, restructuring is subject to a morphological locality condition. Saab claims that the relevant condition is immediate locality (i.e., the relation between a head and the head of its complement; see Embick and Noyer 2001):

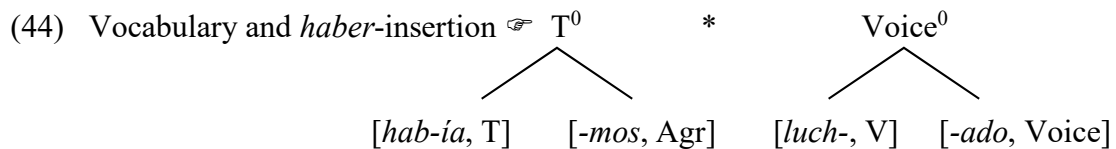
(42) Contextual condition: restructuring at PF requires immediate locality between T^0 and VoiceP.

Then, for the periphrastic form at hand, restructuring combines T^0 and V^0 under the phrasal T' node:

(43) morphological restructuring ➦ TP



Finally, Vocabulary Insertion takes place giving rise to the periphrastic form:



For a synthetic like *luchábamos*, the derivation is the same, although an additional application of Local Dislocation adjoins the inflectional affixes to the verbal base (see Saab 2022a for details).

Three immediate consequences follow from this approach. First, if obligatory restructuring is what is at stake in the formation of simple and complex forms, then the type of proclisis observed in the relevant languages is accounted for as *clitic climbing*.⁴ In effect, as argued by Rizzi (1982), restructuring triggers clitic climbing with well-known cases of restructuring predicates (see the next section). The difference between classical restructuring predicates, in which the rule is optional, and verb formation is that restructuring is obligatory in verb formation, because of a selectional requirement of the T^0 node. Consequently, proclisis in verb formation is forced compulsively both in synthetic and analytic tenses:

- (45) a. *Lo* hicimos.
 CL did
 ‘We did it.’
 b. *Lo* habíamos hecho.
 CL had.1PL done
 ‘We had done it.’

Second, the approach explains why languages with generalized restructuring do not have verbal ellipsis. Concretely, restructuring, defined as a PF operation, is bled by syntactic ellipsis, i.e., ellipsis bleeds restructuring as another case of the Elmo in (18), repeated below:

- (46) Ellipsis-Morphology Generalization (Elmo): For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

Recall the relevant ellipsis facts in Spanish, French, Catalan or Italian in (32)-(35). Below I repeat the Spanish examples:

- (47) Aux-stranding VP-ellipsis: no
 a. A: Habías comprado el libro?
 had bought.2SG the book
 ‘Had you bought the book?’
 B: * Sí, HABÍA.
 yes had.1sg
 Intended: ‘Yes, I had.’
 V-stranding VP-ellipsis: no

⁴ This does not imply that all varieties of proclisis across languages require restructuring. For sure this cannot be the case in European Portuguese, which has proclisis in negative clauses but still licenses V-stranding VP-ellipsis in negative clauses (see Martins 1994: 188-189), and cannot be the case in Serbo-Croatian either, which, as we will see in section 5, has *clitic*-stranding VP-ellipsis.

- B: * Sí, DI.
yes gave.1SG
Intended: ‘Yes, I did.’ (lit. ‘Yes, I gave.’)

Finally, the theory not only predicts absence of VP-ellipsis in restructuring languages but also absence of syntactic VP-fronting, since phrasal movement of the relevant verbal category also bleeds the context for morphological restructuring. Note first that, in Portuguese, Galician or English, this type of fronting is possible:

- c. ... e estudado, eu *tinha* <[VoiceP estudado]>
and studied I had studied
'and studied, I had.'

(49) a. * Dije que iba a estudiar de nuevo
said.1SG that went.1SG to study.INF again
aunque [vP estudiado]_i, yo había t_i.
although studied I had

- Intended:* ‘I said I was going to study again, although study, I had.’
- b. * Anne voulait manger des moules et
 Anne wanted eat.INF some mussels and
 [_{VP} mangé des moules]_i elle a _{t_i}.
 eaten some mussels she has
- Intended:* ‘Anne wanted to eat mussels, and eat mussels, she did.’
- [Authier 2011:213]
- c. *? [_{VP} Comprato il libro]_i, aveva _{t_i}.
 bought the book had.3SG
- Intended:* ‘Buy the book, (s)he had.’

- (50) Generalized restructuring languages:
- a. uniform proclisis in finite tenses
 - b. lack of syntactic autonomy of the VP, i.e., VP-frozenness effects

I then conclude that the set of correlations attested between clitic placement, ellipsis and movement gives particular support for the model of the timing of ellipsis and morphology illustrated below:

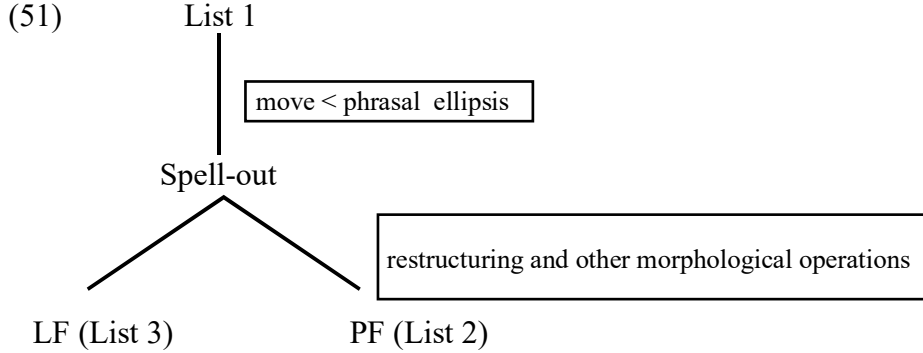


Figure 2. *Timing and interactions between ellipsis and other grammatical operations*

In the next section, I review an additional set of facts in Spanish and Hungarian that independently supports the timing approach sketched in this section.

4. More on restructuring and ellipsis

In this section, I address two empirical domains that further illustrate the interaction between restructuring and ellipsis in connection to the Elmo, repeated below:

- (52) Ellipsis-Morphology Generalization (Elmo): For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

First, I focus on ellipsis with canonical restructuring predicates in Spanish and their interaction with clitic climbing. As I will show, Spanish and related Romance languages have ellipsis with optional restructuring predicates only when restructuring does not apply, a fact that provides independent evidence in favor of the approach to the verbal ellipsis parameter discussed in the previous section. Second, I review the analysis of particle reduplication in Hungarian offered in Lipták and Saab (2019). Hungarian is interesting because it has another variety of verbal ellipsis, concretely, *particle*-stranding VP-ellipsis. The language also has particle reduplication, but crucially it does not have *particle-particle-stranding VP-ellipsis*. As we will see, this curious fact follows again from the Elmo and the clitic behavior of Hungarian particles.

4.1. Clitic climbing, restructuring and modal ellipsis in Spanish

Although Spanish does not have verbal ellipsis of the type that productively occurs in Portuguese or English, it does have null predicate complements with certain verbs traditionally cataloged within the class of restructuring verbs, like *poder* ‘can’ or *querer* ‘to want’:

- (53) a. Ana pudo hacerlo y Paula también pudo.
 Ana could do.INF-CL and Paula also could
 ‘Ana was able to do it, and Paula was able too.’

- b. Ana quiere ir al cine y Paula también quiere.
 Ana wants go.INF to.the movies and Paula also wants
 ‘Ana wants to go to the movies and Paula also wants to.’

The nature of the silence in the sentences in (53) is controversial, although the received view, in particular, since Depiante (2000, 2001), is that they are null anaphora complements (NCA), that is, some kind of deep anaphora. Depiante’s analysis has roughly this form, in which $\emptyset$ stands for a null proform:

- (54) Juan pudo hacerlo y María también pudo $\emptyset$.
 Juan could do.INF-CL and María also could
 ‘Juan was able to do it and María also was able to.’

The strongest evidence that Depiante uses in favor of this analysis has to do with two effects related to the restructuring process. The first relevant fact is that, although (54) is perfectly licit, any attempt to raise the clitic in the second conjunct is frankly ungrammatical, regardless of the presence or absence of restructuring in the first conjunct:

- (55) a. * Juan pudo hacerlo y María también lo pudo.
 Juan could do.INF-CL and María also CL could
 b. Ana lo quiso hacer, aunque finalmente no (*lo)
 Ana CL wanted do.INF although finally not CL
 pudo.
 could
 ‘Ana wanted to do it, although finally she was not able to.’

The second fact is related to the impossibility of having NCA in cases of object preposing. For Italian, Rizzi (1982) already observed that restructuring allows for the object of the infinitive to agree with the restructuring verb. The same happens in Spanish in contexts of the so-called passive *se* construction:

- (56) a. Se puede(n) vender estos libros fácilmente.
 SE can.(PL) sell.INF these books easily
 b. Estos libros se pueden vender fácilmente.
 these books SE can.PL sell.INF easily
 ‘These books can be sold easily.’

Note that in postverbal position the plural object *estos libros* may or may not induce agreement with the modal verb, but, in cases of object preposing, agreement is mandatory, as shown by the contrast between (56) and the following ungrammatical sentence:

- (57) * Estos libros se puede vender fácilmente.
 these books SE can.SG sell.INF easily

At any rate, agreement with the modal indicates restructuring, obligatory in the case of object preposing or optional with canonical postverbal objects. Now, Depiante notes that, in object preposing contexts, NCA is illicit:

- (58) * Estos libros se pueden vender fácilmente y

these	books	SE	can.PL	sell.INF	easily	and
estas	revistas		también	se	pueden.	
these	magazines		also	SE	can.PL	

According to Depiante’s account, the two aforementioned effects are the byproduct of the atomic nature of null anaphoric complements, which lack internal structure and, consequently, cannot contain either clitics or internal objects to extract. In effect, if the silent complements of certain restructuring verbs were cases of ellipsis, we would expect that both clitic climbing and object preposing would be as licit operations as they are in the non-silent counterparts of the same sentences. Depiante’s reasoning is impeccable, but incomplete. Indeed, Saab (2008) showed that other types of extractions are perfectly licit with modal verbs such as *poder* or *querer* (see also Fernández-Sánchez 2023 for examples of *wh*-extraction):

- (59) a. La estudiante a la que quisiste desaprobar
the student DOM the that wanted.2SG fail.INF
y la estudiante [a la que no pudiste
and the student DOM the that not could.2SG
< desaprobar t>] presentaron ambas una queja.
fail.INF filed both a complaint

‘The student who you wanted to fail and the student who you were not able to fail both a complaint.’

- b. A Ana, puedo desaprobarla, pero [a Paula
DOM Ana can.1SG fail.INF-CL.ACC.3SG but DOM Paula
no puedo < desaprobarla >]
not can.1SG fail.INF-CL.ACC.3SG
‘Ana, I can fail her, but Paula, I cannot.’

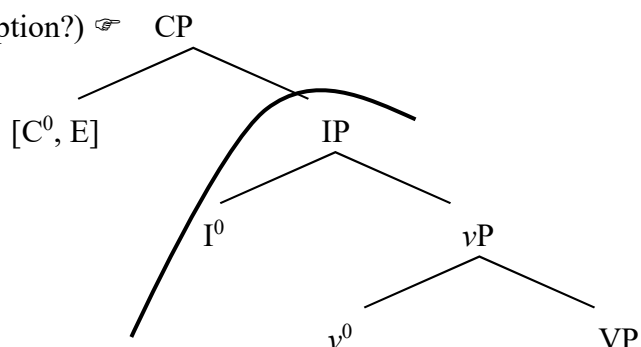
The first example is a case of relativization by means of an explicit relative operator like *la que*. The second is a case of clitic-left dislocation. Both the relative operator and the dislocated topic are differentially marked with *a*. By uniformity considerations, such differential marking must be obtained in the same way it obtains in the non-elliptical counterparts of the same sentences. The bottom line is that the E-sites in (59) must contain underlying structures and cannot be treated as atomic proforms. Put differently, it has to be the case that we are dealing with cases of modal-stranding XP-ellipsis.

This is the same conclusion that Dagnac (2010) and, very recently, Fernández-Sánchez (2023) reach. In addition, Fernández-Sánchez claims that examples such as those in (55a) are ungrammatical for the simple reason that Spanish does not have VP-ellipsis. His key assumption is that the difference between restructuring and non-restructuring predicates is not in a Rizzi-style rule, but rather in a “size difference”, as argued originally by Wurmbrand (1998). In effect, Wurmbrand argued that the effects of Rizzi’s rule can be understood in terms of different projection levels in the functional spine of nonfinite complements. In her terms, restructuring verbs are ambiguous between a structure that contains at least one TP or CP projection and a smaller structure that only projects one *vP* or VP.

- (60) a. Non-restructuring complements: [*poder* [_{CP/TP} ... INF ...]]
b. Restructuring complements: [*poder* [_{vP/VP} ... INF ...]]

Greater or lesser functional structure would correspond, then, to other typical effects traditionally attributed to Rizzi's restructuring rule. For example, clitic climbing or object preposing are only licit in configurations in which no sentence boundary intervenes, which is the case in (60b). According to Fernández-Sánchez, these are precisely the configurations in which Spanish does not have modal ellipsis. In other words, the author's explanation belongs to that family of lexicalist theories of ellipsis licensing, according to which whether a phrase of any category type can be elided depends essentially on some idiosyncratic lexical property of the [E]-bearing head. From this point of view, Fernández-Sánchez ascribes to the idea outlined in the tree of (61):

(61) High ellipsis in Spanish (¿only option?) ➡



More specifically, Spanish only admits ellipsis of an inflectional phrase, perhaps TP, or even some higher category. The problem, in this case, is that a lexicalist explanation such as the one that Fernández-Sánchez suggests does not allow for capturing the entire set of properties listed in (50). For example, a Romance language with restructuring and Aux- or V-stranding XP-ellipsis of the type productively attested in different dialects of Portuguese and Galician would be quite to be expected, contrary to the facts. On the explanation offered in the previous section, the type of proclisis observed in the set of relevant finite forms in the languages listed in (32)-(35) is a robust indication of morphological restructuring. Note that, even though it is not an explanation of the correlation, it does at least allow a sufficiently robust heuristic determination criterion. The idea is just to assume that in a sentence like

- (62) Ana lo reseñó.
 Ana CL.ACC.MASC.3SG reviewed
 ‘Ana reviewed it.’

proclisis is produced by whatever is the reason that produces sentences like

- (63) Ana lo quiso/pudo reseñar.
 Ana CL.ACC.MASC.3SG wanted/could review.INF
 ‘Ana wanted/was able to review it.’

That it is to say, proclisis of the Spanish-type requires morphological merger to a restructured predicate. The only difference between (62) and (63) is that in (62) proclisis is mandatory, i.e., in the relevant Romance languages sentences such as:

- (64) * Ana reseñólo.
 Ana reviewed- CL.ACC.MASC.3SG

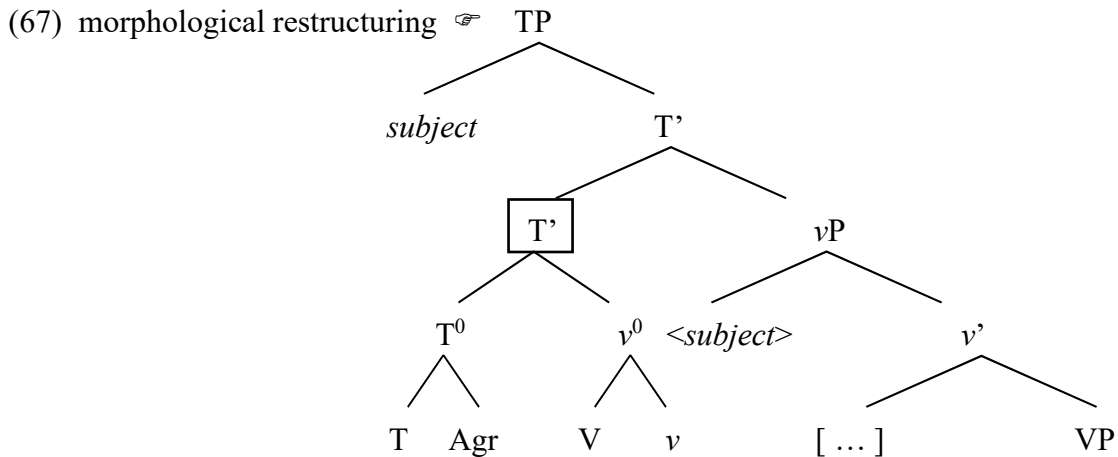
are illicit even if enclisis is perfectly licit as an alternative for (63):

- (65) Ana quiso/pudo reseñar lo.
 Ana wanted/could review.INF-CL.ACC.MASC.3SG
 ‘Ana wanted/was able to review it.’

Having this background in mind, let us see, then, what is wrong with sentences such as (55b), repeated below, in which there is proclisis under Modal-stranding ellipsis:

- (66) * Ana lo quiso hacer, aunque finalmente no lo
 Ana CL wanted do.INF although finally not CL
pudo < hacer t_{lo} >.
 could do.INF

The cause of the ungrammaticality of this example should now be transparent: proclisis is simply impossible in the absence of a previous restructuring step, like the one we tentatively suggested in (43), repeated again for the convenience of the reader:



Effectively, in the case of (66), the infinitive in the E-site and the modal verb cannot be subject to restructuring due to an infraction of the Elmo. The result is, then, that the clitic cannot be morphologically adjoined as a proclitic. If there is no restructuring -which, as we saw, is “optional” with certain predicates-, then ellipsis of the infinitival complement is perfectly licit:

- (68) Ana lo quiso hacer, aunque finalmente no pudo
 Ana CL wanted do.INF although finally not could
 < hacerlo >.
 do.INF-CL
 ‘Ana wanted to do it, although finally she was not able to.’

A similar explanation can be extended to the incompatibility between object preposing and the modal ellipsis illustrated in (58). Again: a restructuring environment is a precondition for object preposing to apply successfully. Yet, by the Elmo, ellipsis blocks restructuring, rendering any attempt of extracting the object illicit. These considerations lead us to what is perhaps another important correlation of the verbal ellipsis parameter, namely:

- (69) Romance languages with generalized restructuring have modal ellipsis with some verbs that show optional restructuring, only in case restructuring does not apply.

As we have already seen, the extraction data in (59) show that, in fact, Spanish has head stranding ellipsis with certain optional restructuring predicates (see Fernández-Sánchez 2023 for more evidence). The theory of generalized restructuring sketched in the previous pages resolves the paradox involving apparently contradictory evidence for detecting true deletion in Modal-stranding ellipsis. It essentially reduces the impossibility of clitic climbing and object preposing in contexts of modal ellipsis to the bleeding effects that ellipsis produces on restructuring.

4.2. *VP-ellipsis and particle reduplication in Hungarian*

So far, I have argued that Spanish allows for certain stranding ellipses in very restricted scenarios involving modal verbs or related restructuring predicates. In effect, as we have seen in section 4.1, enclisis vs. proclisis in canonical restructuring environments correlates with presence or absence of ellipsis, respectively. To repeat: the reason why (66) is ungrammatical follows from the Elmo in (52). In this particular case, what ellipsis bleeds is restructuring. Unlike synthetic or analytical tenses, canonical restructuring verbs have a broader distribution, allowing for different complement sizes. Whenever they take TPs or higher projections as complements, two things happen: (i) there is no restructuring, and (ii) ellipsis of the infinitival complement is now licensed, as in (68). These facts, and the theory that accounts for them, have an implication of great importance for the verbal ellipsis parameter, namely, we cannot think of it as a matter of *yes/no* specification over an abstract universal property (of the classical type: *VP-ellipsis* [*yes/no*]), but we must attend to particular grammatical factors that must be determined on the basis of language-internal properties, in occasions, in a case-by-case manner. To illustrate this point beyond Spanish, consider the case of Hungarian, a language which, unlike Spanish, has Aux-stranding ellipsis (see (70)), V-stranding ellipsis (see (71)), and Particle-stranding ellipsis (see (72)):

- (70) Aux-Stranding ellipsis:
- | | | | | | |
|----|--|-------------|--------------|----|-----------|
| a. | Péter alszik, | és én | is fogok | < | aludni >. |
| | Péter sleep.3SG | and I | also FUT.1SG | | sleep.INF |
| | ‘Péter is sleeping and I will, too.’ | | | | |
| b. | Péter alszik, | de nekem | nem kell | < | aludni>. |
| | Péter sleep.3SG | but 1SG.DAT | not need | | sleep.INF |
| | ‘Péter is sleeping but I don’t need to.’ | | | | |
| c. | Többet aludtam | ma, mint | amennyit | | máskor |
| | more slept.1SG | today than | how.much.ACC | | otherwise |
| | szoktam | < | aludni | >. | |
| | HABIT.1SG | | sleep.INF | | |
| | ‘I have slept more today than I usually do on other days.’ | | | | |
- [Lipták 2019: 821]

- (71) V-stranding ellipsis:
- | | | | |
|----|---------------------------------------|---------|-----------------------|
| A. | Fel hívta | Bea a | szüleit |
| | PV called.3SG | Bea the | parent.POSS3SG.PL.ACC |
| | tegnap? | | |
| | yesterday | | |
| | ‘Did Bea call her parents yesterday?’ | | |

B. Fel hívta.
PV called.3SG

[Lipták 2019: 823]

(72) Particle-stranding ellipsis:

A. Fel hívta Bea a szüleit
PV called.3SG Bea the parent.POSS3SG.PL.ACC
tegnap?
yesterday
'Did Bea call her parents yesterday?'

B. Fel.
PV
'She did.'

[Lipták 2019: 825]

Yet, the language does not have [*particle particle*]-stranding ellipsis, even when it *does* have particle reduplication (see (73) and Lipták and Saab (2019) for extensive discussion and references) and, as we have just seen, stranding ellipsis of various sorts. Thus, leaving the reduplicated particles stranded leads to ungrammaticality (see (73B)). In such a scenario, Part-Part-V-stranding ellipsis is, instead, a grammatical alternative (see (73B')):

(73) No Reduplicated Particle-stranding Ellipsis:

A: Be-be kukkant azért a nagyszülőkhöz Peti
IN-IN peep.3SG still the grandparent.PL.ALL Peti
néha?
sometimes
'Does Peti visit his grandparents sometimes?'

B: * Be-be.
IN-IN

B': Be-be kukkant.
IN-IN peep.3SG
'He does.'

[Lipták and Saab 2019: 540]

This ban of Part-Part-stranding ellipsis correlates with the lack of syntactic autonomy of reduplicated particles in general. Roughly speaking, Hungarian simple particles have a hybrid clitic-like behavior, behaving both as phrases and heads. In contrast, reduplicated particles have an affix-like behavior, i.e., they are rigid subwords. Among other relevant facts, Lipták and Saab show that, unlike regular part-verb constructions (see (74)), particle reduplication does not allow for contrastive focus on the reduplicated particles (see (75)):

(74) A: BE nézett az ablakon?
IN look.PST.3SG the window.SUP
'Did he look IN the window?'

B: Nem. KI nézett.
no OUT look.PST.3SG
'No. He looked OUT the window.'

[Lipták and Saab 2019: 538]

(75) A: * BE-BE nézett az ablakon?
IN-IN look.PST.3SG the window.SUP
'Did he look IN the window?'

- B: Nem. *KI-KI nézett.
 no OUT-OUT look.PST.3SG
 ‘No. He looked OUT the window.’

[Lipták and Saab 2019: 538]

In the same vein, left dislocation of the reduplicated particles in verum focus constructions is disallowed, as the following contrast with regular part-verb constructions shows (small capitals on the verb indicates verum focus):

- (76) a. Ki, NÉZTEM.
 OUT look.PST.1SG
 lit. ‘Out, I did look.’
 b. * Ki-ki, NÉZTEM.
 OUT-OUT look.PST.1SG
 lit. ‘Out, I did look from time to time.’

[Lipták and Saab 2019: 539]

Finally, reduplicated particles cannot be inverted in contexts requiring particle inversion (typically, sentences with preverbal focus as in (77b) or negation as in (77c-d)), i.e., there is a left-adjacency requirement for reduplicated particles, absent with non-reduplicated ones, which are grammatical in exactly the same environments.

- (77) a. Peti bele-bele nézett a könyvbe.
 Peti INTO.3SG-INTO.3SG look.PST.3SG the book.ILL
 ‘Peti looked into the book from time to time.’
 b. * PETI nézett bele-bele a könyvbe.
 Peti look.PST.3SG INTO.3SG-INTO.3SG the book.ILL
 ‘It was Peti who looked into the book from time to time.’
 c. * Nem nézett bele-bele a könyvbe.
 not look.PST.3SG INTO.3SG-INTO.3SG the book.ILL
 ‘He did not look into the book from time to time.’
 d. * A kismackó nem állt meg-meg az
 the little.bear not stop.PST.3SG PRF-PRF the
 erdőben.
 woods.IN
 ‘Little bear did not stop occasionally in the woods.’

[Lipták and Saab 2019: 534]

This type of evidence points towards the quite robust conclusion that particle reduplication requires some kind of morphological restructuring by virtue of which the reduplicated particle and the verb form a complex head. More concretely, Lipták and Saab conjecture that particle reduplication requires a previous instance of morphological reanalysis between the particle and the verb:

- (78) *Conjecture:* Particle reduplication is possible iff reanalysis has formed a complex morphosyntactic word containing the verb and the particle.

[Lipták and Saab 2019: 548]

In a way similar to clitic climbing under restructuring, Hungarian has reduplication under a process of morphological restructuring, which might be roughly decomposed in two

basic steps: (i) particle and V^0 restructuring, and (ii) reduplication within the restructured word.

(79) [Part [x^0 Verb ... $\rightarrow$ [... [x^0 Part+Verb] ... $\rightarrow$ [... [x^0 Part [x Part-Verb]] ...

Thus, any attempt to leave reduplicated particles stranded like in (73B) will destroy the constituency of X^0 . The solution is, then, (73B'), in which the stranded word is the complex MWd containing the verb and the reduplicated particles.

5. Ellipsis and second-position clitics in Serbo-Croatian

The ellipsis facts in Romance and Hungarian robustly point to the need for a PF movement approach in the derivation of clitics/particles. Put differently, syntax alone cannot resolve clitic placement; further morphological operations have also an impact in the clitic/particle derivations. The argument is based on the premise that phrasal ellipsis applies before morphological operations (see Figure 2 in (51)), with the “dramatic” consequence of bleeding them under specific circumstances. In this section, I review a similar argument made by Stjepanović (1999) in the domain of second-position cliticization and clitic clustering in Serbo-Croatian (SC). Recall the example in (5) from the introduction, which illustrates the canonical second position for clitics in SC:

- (80) a. [Tu knjigu] su mi dali.
 that book.ACC are me-DAT given
 ‘They gave that book to me.’
 b. [Tu] su mi knjigu dali.
 that are me-DAT book.ACC given
 ‘They gave that book to me.’

[Stjepanović 1999: 14]

Importantly, sequences of clitics form a cluster that cannot be interrupted by other constituents and, in the vast majority of morphophonological environments, these clitic sequences respect the following linear order (*li* = interrogative clitic, *je* = third person auxiliary):

- (81) *li*-AUX-DAT-ACC-GEN-REFL-*je*

[Stjepanović 1999: 14]

As already mentioned, there is a dispute between *Syntax Only* theories and *PF movement* ones, according to which syntax alone cannot do the entire job in the derivation of clitics (see section 1 and Embick and Noyer 2001). As for pure syntactic views, Progovac (1996) presents a particular clear analysis both of second-position effects and clitic clustering. Let me illustrate her theory considering her analysis for the following clitic cluster in second position (clitics are underlined):

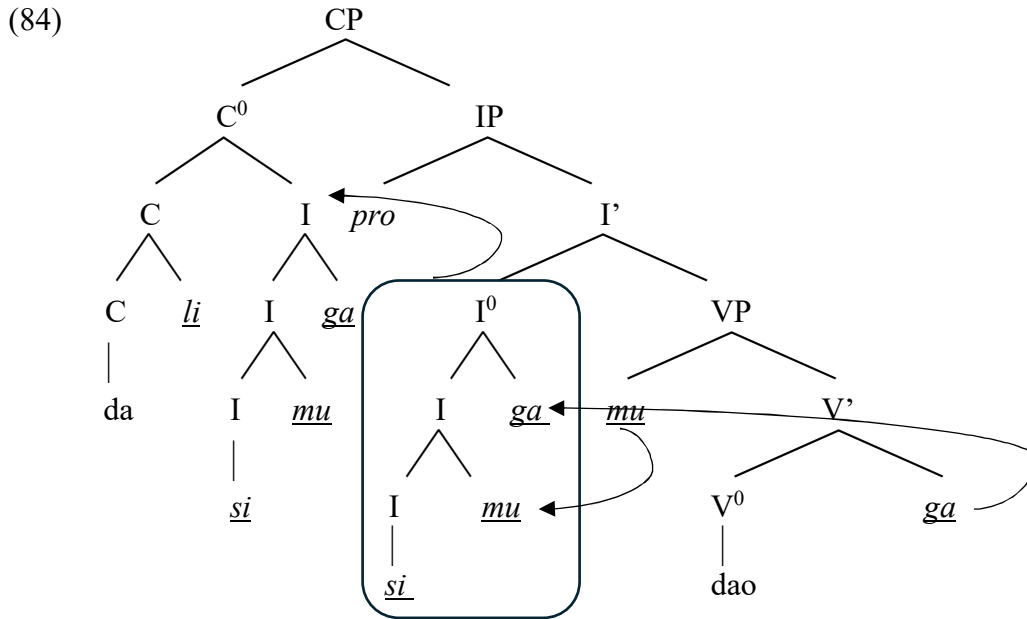
- (82) Da=li=si=mu=ga dao?
 that=Q=are=him=it given
 ‘Have you given it to him?’

[Progovac 1996: 420]

The core ingredients of Progovac’s analysis are listed below:

- (83) a. Clitics in SC are right-adjoined to Comp, i.e., second position *is* Comp;
 b. Clitics in SC move to Comp in syntax:
 c. Only elements that can move to Spec of CP or Comp, or are base-generated in Comp can support clitics: First position is either Specifier of CP or C(omp).
 [Progovac 1996: 412]

For a case like (82), she assumes that clustering also proceeds entirely in the syntax, with the dative clitic adjoining to I^0 before the accusative clitic. The complex [I^0 *si*-DAT-ACC] then attaches to the C^0 head, which already has another clitic attached, namely, the question particle *li*. The simplified tree in (84) illustrates what I have just said:



[adapted from Progovac 1996: 421, (41)]

Other Syntax Only approaches to Slavic languages are Rivero (1991, 1999) and Cavar and Wilder (1994). Differences among purely syntactic proposals have to do with the motivation for clustering and second position, among other things. At any rate, for all of them Stjepanović presents an important challenge coming from ellipsis. Let me review her argument with some detail. First, note that SC makes use of Aux-Stranding VP-ellipsis. In the following example, the clitic auxiliary *ste* gets stranded after verbal ellipsis:

- (85) ? Oni su kupili novine, a i vi ste
 they are bought newspapers, and also you are
 < kupili novine > (takodje).
 bought newspaper too
 'They bought the newspapers, and you did, too.'

[Stjepanović 1999: 17]

Consider next the following minimal pair in which both Aux-Stranding and Aux-DAT-Stranding VP-ellipsis is allowed. In effect, note that in (86b), the dative clitic gets stranded after ellipsis with independence of the accusative clitic, which remains within the E-site:

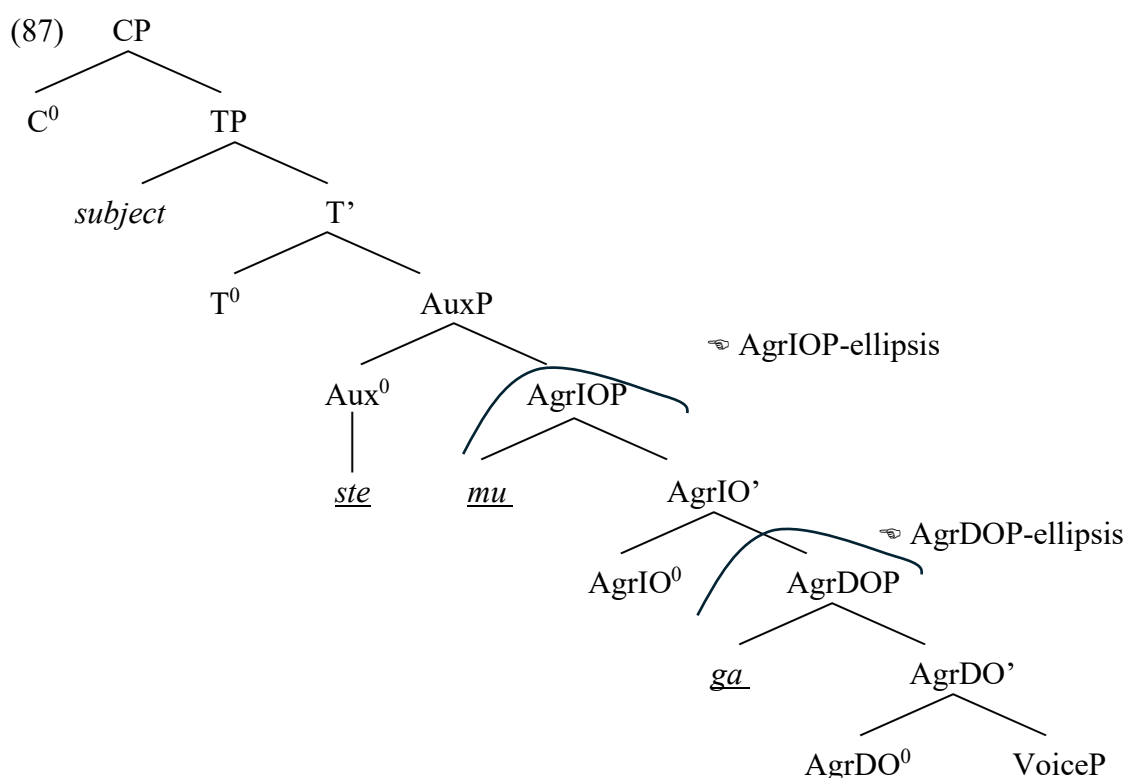
- (86) a. ?Mi *smo* *mu* *ga* predstavili, a i
 we are him-DAT him-ACC introduced and also
 vi *ste* < *mu* *ga* predstavili >, (takodje).
 you are him-DAT him-ACC introduced too
 ‘We introduced him to him, and you did, too.’

[Stjepanović 1999: 29]

- b. ?Mi *smo* *mu* *ga* predstavili, a i
 we are him-DAT it-ACC introduced and also
 vi *ste* *mu* < *ga* predstavili >, (takodje).
 you are him-DAT it-ACC introduced too
 ‘We introduced him to him, and you did, too.’

[Stjepanović 1999: 30]

This type of facts strongly suggests that clitics are syntactically merged in different maximal projections. Then, different ellipsis sizes would explain the distribution of clitic remnants. For instance, for a case like (86a), in which only the auxiliary clitic survives, Stjepanović assumes that the E-site includes all the clitic projections in the middle field (AgrIOP and AgrDOP). In turn, a smaller constituent is elided in (86b), perhaps AgrDOP and all the projections it dominates. In sum, depending on the size of the E-site, we would have only the auxiliary as remnant or the auxiliary plus the dative clitic. The two relevant E-sites are illustrated below:



Based on these considerations involving ellipsis and clitic placement, Stjepanović concludes the following:

“The fact that clitics are found in distinct maximal projections is difficult for syntactic approaches to account for. As shown above, in these accounts, clitics crucially must be adjoined to each other or to the same node in overt syntax. Leaving only one of them

within an ellipsis site then should not be possible. If it were possible not to adjoin clitics to each other or to the same node, leaving them in separate maximal projections in VP ellipsis constructions, the same situation could hold in an equivalent sentence without VP ellipsis.”

[Stjepanović 1999: 31]

Now, if clustering is the byproduct of a morphological rule applying after syntax, then the fact that in (86b) the accusative clitic does not form a cluster with the clitic remnants follows as another instance of the Elmo, repeated below:

- (88) *Ellipsis-Morphology Generalization (Elmo)*: For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

In (86b) the accusative clitic remains within the E-site and, consequently, cannot move to form a cluster with the surviving clitics. Importantly, SC also has smaller VP-ellipses with all clitics surviving the E-site. According to Stjepanović, this ellipsis variety requires strong auxiliaries in order to get proper contrast between the auxiliary in the antecedent and the remnant in the clause containing the E-site. The following examples show this pattern in which both pronominal clitics survive the E-sites:

- (89) a. Ja *sam* *mu* *ga* dala,
 I am him-DAT it-ACC given,
 a ti *mu* *ga* nisi < dala >.
 and you him-DAT it-ACC aren't given
 'I gave it to him, but you didn't give it to him.'
- [Stjepanović 1999: 37]
- b. Ja *mu* *ga* nisam predstavila,
 I him-DAT him-ACC am-not introduced,
 a ti *mu* *ga* jesi < predstavila >.
 but you him-DAT him-ACC are introduced
 'I didn't introduce him to him, but you did.'
- [Stjepanović 1999: 41]

This fact shows that one cannot attribute clitic placement in (86) to some putative ban for accusative clitic extraction under ellipsis. In principle, all pronominal clitics can be ellipsis remnants, but this entirely depends on the ellipsis size. In any case, note that whenever the clitics survive the E-site, then they form clusters in the usual way. I will not discuss any concrete implementation of the PF theory of second-position and clitic clustering. Among other relevant studies, PF-movement approaches can be found in Halpern (1995) and Embick and Izvorski (1997). Bošković (2001) is an important alternative, according to which clustering and second position in clitic constructions follow from PF considerations, as well, but not from PF movements.

Before closing this section, I would like to mention another fact in SC which connects *je* placement to the Elmo, providing thus further evidence for PF approaches to clitic placement. Recall that this auxiliary clitic occurs at the final position of the cluster, following all pronominal clitics (see (82)):

- (90) On mi ga je dao.
 he me-DAT it-ACC is given
 'He gave it to me.'

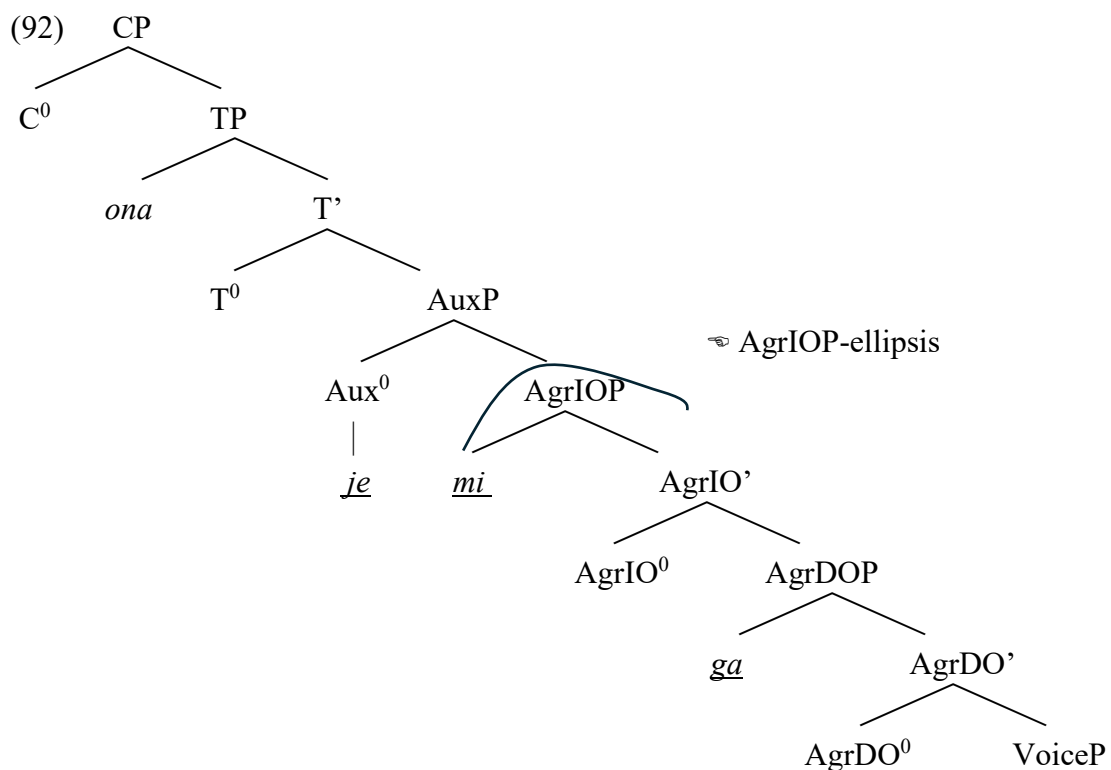
[Stjepanović 1999: 55]

Yet, under AgrIOP-ellipsis, *je* survives, showing that it has to be located in Aux in the syntax:

- (91) a. ? On *mi* *ga* *je* predstavio,
 he me-DAT him-ACC is introduced,
 a i ona je (takodje).
 and too she is as well
 'He introduced him to me, and she did, too.'
- b. * On *mi* *ga* *je* predstavio,
 he me-DAT him-ACC is introduced,
 a i ona mi ga, (takodje).
 and too she me-DAT him-ACC (too)
 'He introduced him to me, and she did too.'

[Stjepanović 1999: 56]

The following tree illustrates the syntactic ordering between *je* and the pronominal clitics deduced from ellipsis:



From the Elmo perspective, this sort of ordering puzzle vanishes. Concretely, in non-elliptical *je* is syntactically higher than pronominal clitics; then, at PF, *je* lowers into the clitic cluster surfacing at the right edge. Yet, under ellipsis, as we already know, lowering is prevented by the Elmo (see Stjepanović 1999 and Bošković 2001 for other PF alternatives to *je* placement).

6. Summary and further clitic/ellipsis interactions

As noted in the introduction, language after language clitics have many perplexing properties. Two families of theories have been under dispute in the last three decades in the generative framework. For some, syntax is enough to account for all the set of intricate properties special clitics show. Yet, for others, syntax alone cannot account for the distribution of clitics and further action is postulated after syntax. The case studies discussed in the preceding sections favor this latter type of approaches. The general argument developed throughout this chapter is based on the idea that ellipsis blocks certain morphological operations. This was stated as the Elmo generalization:

- (93) *Ellipsis-Morphology Generalization (Elmo)*: For every morphological operation MO that affects the domain of X, where X contains the target of MO, MO cannot apply in X if X is subject to ellipsis.

The absence of VP-ellipsis in uniform proclitic languages, the behavior of modal ellipsis with restructuring predicates in Spanish and the ban of *part-part*-stranding VP-ellipsis in Hungarian point out to the conclusion that ellipsis intervenes before certain morphological processes. In turn, different patterns of clitic-stranding ellipsis in Serbo-Croatian show that both auxiliary and pronominal clitics are merged in independent functional projections with clitic clustering applying after syntax. In short, clitics cannot be entirely derived in the syntax; further morphological operations crucially contribute to both clitic placement and clitic clustering.

Beyond the Elmo, there are many other interesting interactions between ellipsis and clitics which I would like to briefly mention. Recall that according to the model for the timing of ellipsis sketched in Figure 2, repeated below, there also postsyntactic ellipses, called head ellipsis in Saab (2008, 2022a):

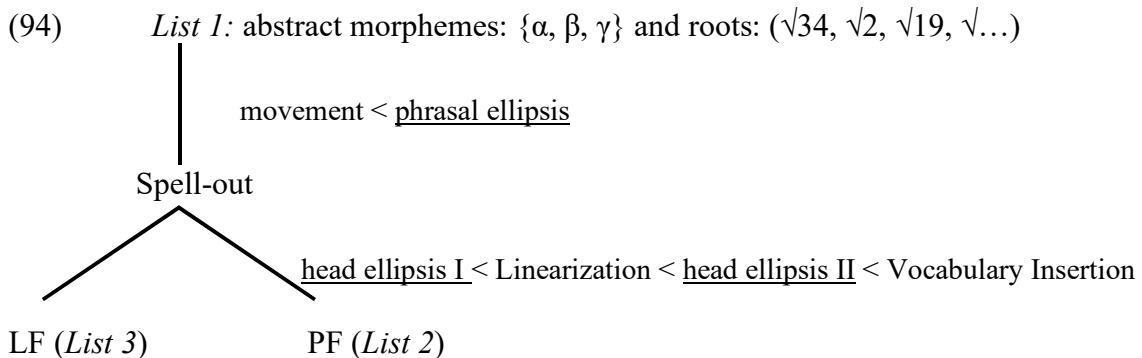
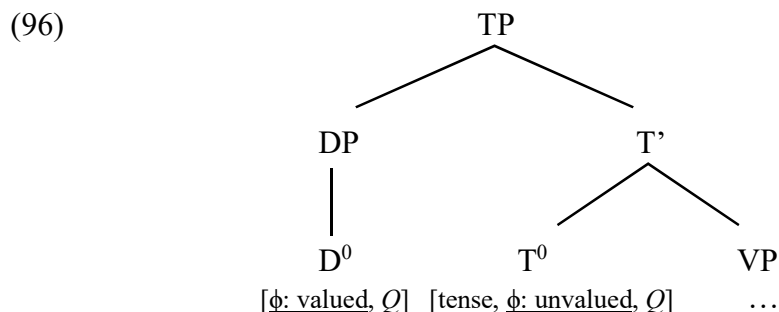


Figure 2. *Derivational ellipsis*

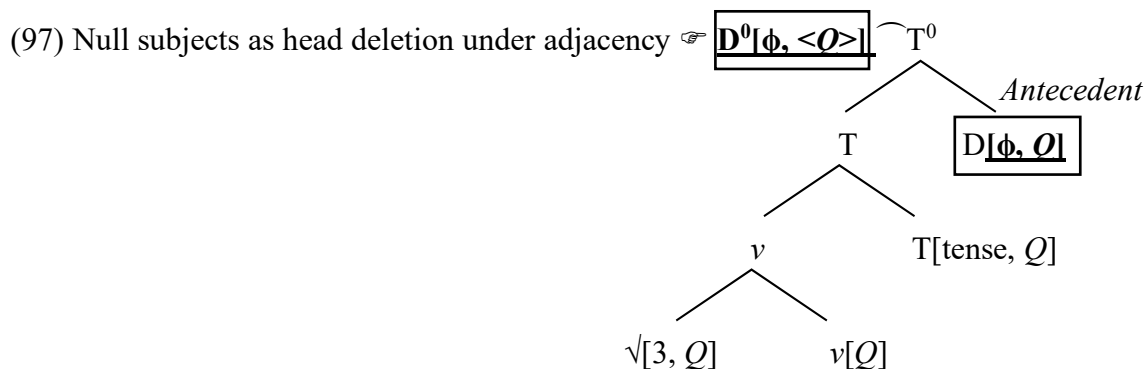
The two varieties of head ellipsis mentioned in Figure 2 are subject to pre- and post-linearization locality conditions. Here is the relevant definition:

- (95) *Head Ellipsis (under Q-deletion)*:
 Given a morphosyntactic word MW, delete every *Q*-feature contained in MW if and only if:
 (i) There is an identical antecedent contained in a morphosyntactic word MW',
 (ii) MW is adjacent or immediately local to MW'

Among the phenomena that are accounted for as instances of head ellipsis, null subjects in consistent null-subject languages are of special interest to the theme of this chapter. According to the ellipsis conception of null subjects proposed in Saab (2008, 2017 and 2024a), a null subject is first generated as a weak pronoun in Spec,TP (weak pronoun in the sense of Cardinaletti and Starke 1999):



At PF, a dissociated morpheme is added to the T^0 node using the inflectional information encoded in the T^0 node itself. Once linearization takes place, the adjacency condition for head ellipsis II under Q -deletion is met (see (95ii)) and the pronominal subject, a morphosyntactic word at PF, is consequently deleted ($\langle \dots \rangle = Q$ -deletion). The derivation for a null subject sentence like *cantamos* ‘we sing’ can be schematized as follows:



Compared to theories that derive null subjects with a lexical primitive like *pro*, this ellipsis approach has a clear advantage since it predicts consistent null subject languages with some obligatory clitic subjects. This is the case, for instance, in Central Trentino and other Northern-Italian dialects, which have all the relevant properties that define the null subject linguistic type (e.g., free inversion of referential subjects), but which cannot elide their clitic subjects. In the following set of sentences, we can see that Central Trentino makes use of obligatory clitics in subject position:

- (98) Central Trentino:
- a. Vèñionte?
come.1SG.CL
‘Should I come?’
 - b. Te vègni con mi.
you.CL come with me
‘You come with me.’
 - c. Vègnet con mi?
come-you.CL with me
‘Will you come with me?’

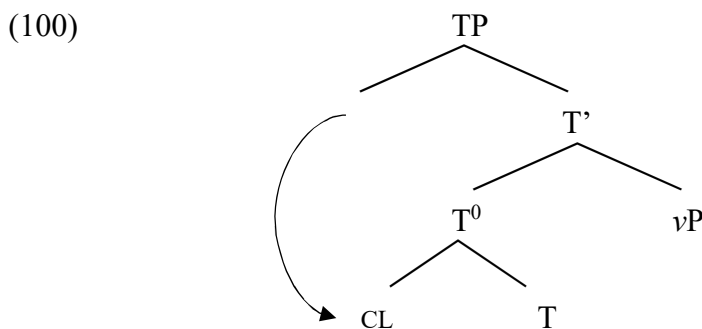
- d. El/la vèn con mi.
 he/she.CL come with me
 ‘He/she comes with me.’
- e. Vègnelo/la con mi?
 comes-he/she.CL with me?
 ‘Will he/she come with me?’
- f. Vegninte con ti?
 come-1PL.CL with you?
 ‘Should we come with you?’
- g. I/le vèn con mi.
 they.CL come with me
 ‘They come with me.’
- h. Vègnei/le con mi?
 come they.CL with me?
 ‘Will they come with me?’

[Casalicchio and Cordin 2020: 86-87]

As the following examples show, these clitic subjects cannot be omitted:

- (99) a. *(I/le) vèn con mi.
 they.CL come with me
 ‘They come with me.’
- b. Vègne*(i/le) con mi?
 come they.CL with me?
 ‘Will they come with me?’
- etc...

Now, here is the gist of the idea: Central Trentino has mandatory clitic subjects because they are literal clitics in Cardinaletti and Starke’s taxonomy, i.e., subwords of the T^0 node at PF, as illustrated in the simplified tree below:

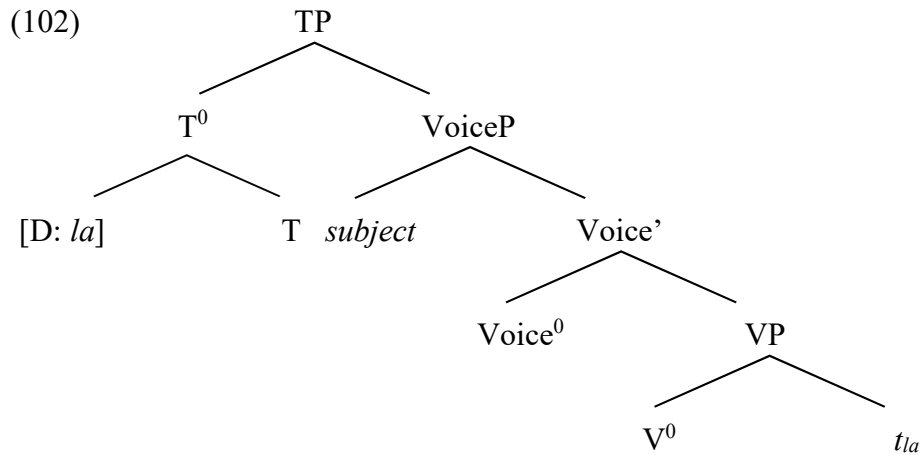


Therefore, the reason why these languages cannot elide clitic subjects is because of the *Subword Deletion Corollary* that prevents ellipsis of subwords adjoined to non-elliptical morphosyntactic words (see section 2.2).

Interestingly, this analysis of clitic subjects in Central Trentino opens important research avenues in other domains involving clitics. For example, as we saw in section 2.1, Matushansky (2006) and others propose that pronominal clitics are generated as phrasal complements and then, move to some position in the functional spine of the clause. Recall

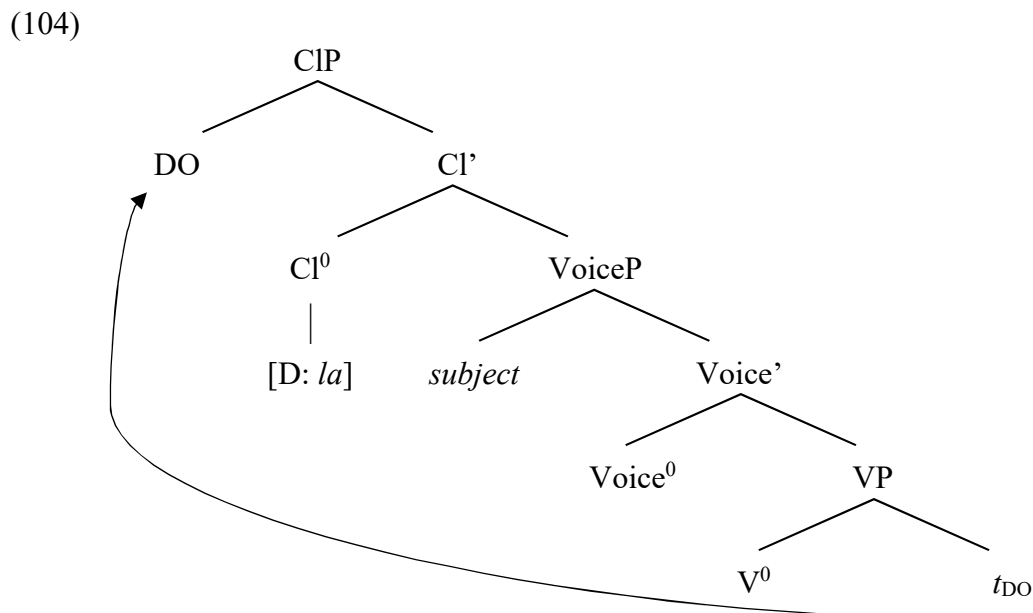
the example in (9) with a simple pronominal clitic and its associated analysis in (11), both repeated below:

- (101) Ana la vio.
 Ana CL.ACC.F.3SG saw
 ‘Ana saw her.’



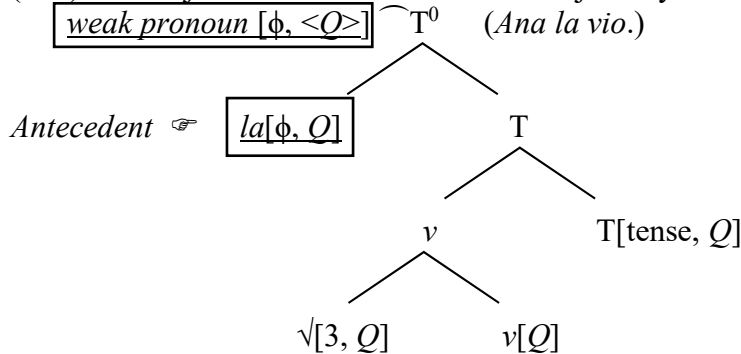
Recall also that I assumed that doubling clitic sentences involve a different derivation according to which the clitic is a probe for A-movement:

- (103) Ana la vio a Paula.
 Ana CL.ACC.F.3SG saw DOM Paula
 ‘Ana saw Paula.’



It is clear now why the ellipsis analysis suggested for Central Trentino makes an alternative approach to (104) very plausible. Concretely, one could assume that those sentences are also doubling structures with weak pronouns elided at PF by an application of head ellipsis. Omitting some important additional steps at morphology, the idea is illustrated in the following tree:

(105) Null objects as head deletion under adjacency:



Importantly, this view is also an alternative to the traditional GB approach according to which clitic structures involves a little *pro* in object position (for Spanish, see Jaeggli 1982). The hope is that further investigations into ellipsis and clitic interactions can further inform us about the best analytic option for some of the many clitic constructions attested in Romance and beyond.

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